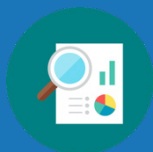


Unit 1 Overview: Basics of Geometry and Constructions

Unit Narrative



The unit is designed to provide students an understanding of many basics of geometry through experiencing transformations and/or constructions. In later units, students will formally use these concepts. The use of transformations and constructions helps students have a “why” to theorems that will be used later.

Students will begin by identifying key words and notations that will be foundational in the comprehension and application of the geometric concepts in the course. Students will explore key words with specific definitions and those words that are accepted as undefined. Once established, students will begin to develop a way of thinking and reasoning that is based on exploration and discovery. A few basic postulates and theorems are developed first through transformations and rigid motions. Students learn to use the tools of geometry to verify transformations and to perform constructions.

The overall flow of this unit is understanding key terms, using these key terms in transformations of figures, and using tools available to do constructions while also using the constructions to understand key terms, geometric ideas, and postulates.

The unit provides several opportunities for teachers to establish norms for different student learning routines as well as opportunities for students to persist in effortful learning and develop a growth mindset.

Unit At-A-Glance

Section 1: Define Key Terms	Benchmarks Addressed	Lesson 1	Defined and Undefined Terms – Part 1
	MA.912.GR.2.2	Lesson 2	Defined and Undefined Terms – Part 2
		Lesson 3	Isometry
		Lesson 4	Dilations and Partitions
Section 2: Transformations	Benchmarks Addressed	Lesson 5	Transformations on the Plane – Part 1
	MA.912.GR.2.3 MA.912.GR.2.4 (<i>Honors only</i>) MA.912.GR.2.6 MA.912.GR.2.8	Lesson 6	Transformations on the Plane – Part 2
		Lesson 7	Congruency and Similarity
		Lesson H1	Symmetries
Section 3: Constructions	Benchmarks Addressed	Lesson 8	Line Segments
	MA.912.GR.5.1 MA.912.GR.5.2	Lesson 9	Copying an Angle
		Lesson 10	Segment Bisectors
		Lesson 11	Perpendicular Bisectors
		Lesson 12	Angle Bisectors



Differentiation

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Engagement <i>The “WHY” of learning</i>	- Multiple opportunities exist to engage students in Math through Art in this unit, especially in the Warm-Ups. In Lesson 3, students watch a video of an Animation Flipbook. Then, connections are made to how the figures on the page were transformed to give the appearance of a cloud moving. Warm-Ups like this recruit learner interest at the beginning of a lesson that will help to sustain engagement throughout the lesson. - Learners should self-regulate their emotions about learning. To develop the ability to do this, students learn about people who have met challenges and how they overcame obstacles. In Lesson 4, students watch an interview with a Disney animator and hear his philosophy on learning. Students reflect on a time in their lives when they had no confidence, but made the first step, and learned from making mistakes.
Representation <i>The “WHAT” of learning</i>	- As students progress into Lesson 2 and beyond, provide a kinesthetic entry-point by having students physically touch and manipulate objects to develop the Geometric concepts that they will need throughout Geometry. This hands-on approach can take many forms and offers multiple points of entry for the students. For example, in Lesson 3, students use patty paper to move and manipulate an object to explore the transformations that are isometries. - Being a foundational unit in Geometry, begin developing a Word Wall. This is first suggested in Lesson 7, however it could be started at Lesson 1 to capture all of the words that are introduced throughout Unit 1.
Action & Expression <i>The “HOW” of learning</i>	- In Lesson 11, students have the option of constructing the perpendicular bisector of a line segment using a compass and paper or using a provided Geogebra applet. Learners should be given the opportunity to use whichever tools might help them fully participate in 1.11.2 Exploration. - In 1.6.2 Guided Instruction, students are asked to complete a chart that can be used to distinguish between similar-looking transformations. Students write hints and then exchange their hints with classmates. This action provides an organizational aid to keep the student’s learning handy for future reference and facilitates the managing of their information.

ELL Information

For students whose first language is not English, vocabulary assistance should be offered for key words throughout the topic. The teacher can highlight the key words and have the student use a dual language dictionary or technology to look up the meaning of the words.

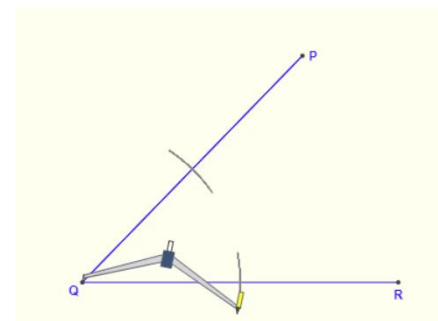
The teacher can print the vocabulary terms large enough to be posted around the room. Using an adhesive, students can post each term next to an object in the classroom that resembles each term.

Lesson 4 focuses parts of a segment. For students whose first language is not English, vocabulary assistance should be offered for keywords. The teacher can highlight and break down the root word of “partition”. The teacher can have the student use a dual language dictionary or technology to look up the meaning of the word.

Students can be directed to the Math Open Reference website to view and review constructions step-by-step.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support” below:

- Congruent • Dilation • Isosceles triangle • Reflection • Right angle • Rigid transformation
- Rotation • Scale factor • Similarity • Straight angle • Translation



Additional Differentiated Support

Scaffolding Resources	On-Ramp
Suggested Instructional Strategies • Explicit instruction on key terms and their meanings will contribute greatly to vocabulary development. • Students may struggle to set up proportions or confuse part-to-part and part-to-whole comparisons. Consider having students re-draw each figure separately or to use highlighters to make comparisons. • Example Activity to copy a segment: have students draw a segment with a straightedge (ruler). Using a dry spaghetti noodle, have students place the noodle above or below the line segment. Students will then place their finger at the endpoint of the line segment on the noodle and break the noodle to create the same length as the line segment. A pipe cleaner may also be used in place of the dry spaghetti noodle by bending the pipe cleaner at the length of the line segment. • Example activity for copying an angle: have students draw an angle using a straight edge (ruler). Using a pipe cleaner, have students copy the angle by bending the pipe cleaner to match the angle. • Example activity to bisect a segment: have students bend a pipe cleaner in half, end to end. Next have students use their thumbs to create a crease in the pipe cleaner. Then have students lay another pipe cleaner on the crease, creating a perpendicular bisector.	Use the videos and practice problems in the On-Ramp to Algebra 1 personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish. • Approximating Irrational Numbers • Plotting Ordered Pairs in the Coordinate Plane Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish. • Horizontal Distance on the Coordinate Plane • Vertical Distance on the Coordinate Plane • Coordinate Points in Real-World Context • Creating Composite Shapes • Lines of Symmetry • Lines, Line Segments, and Rays • Identifying Transformations



Section Overview

Section 1: Define Key Terms (Lessons 1-4)

Benchmarks Addressed	Learning Targets	
MA.912.GR.2.2 Identify transformations that do or do not preserve distance. <i>Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.</i> <i>Clarification 2: Instruction includes recognizing that these transformations preserve angle measure.</i>	Lesson 1	- I can use rigid motions to define parallel lines and perpendicular lines. - I can experiment with rigid motions on a plane. - I can use rigid motions to define a right angle.
	Lesson 2	- I can understand the three undefined terms of geometry. - I can use rigid motions to define straight angles. - I can experiment with rigid motions on a plane.
	Lesson 3	- I can define rigid motions. - I can show rigid motions by mapping using arrows on the plane. - I can explain why isometry preserves distance.
	Lesson 4	- I can dilate a line segment. - I can define midpoint using dilation and understand partitions. - I can relate scale factor to dilation.

Section 2: Transformations (Lessons 5-7 and H1)

Benchmarks Addressed	Learning Targets	
MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure. <i>Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.</i> <i>Clarification 2: Within the Geometry course, figures are limited to triangles and quadrilaterals and rotations are limited to 90°, 180° and 270° counterclockwise or clockwise about the center of rotation.</i> <i>Clarification 3: Instruction includes the understanding that when a figure is mapped onto itself using a reflection, it occurs over a line of symmetry.</i> MA.912.GR.2.4 (Honors only) Determine symmetries of reflection, symmetries of rotation and symmetries of translation of a geometric figure. <i>Clarification 1: Instruction includes determining the order of each symmetry.</i> <i>Clarification 2: Instruction includes the connection between tessellations of the plane and symmetries of translations.</i> MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent. <i>Clarification 1: Instruction includes showing that the corresponding sides and the corresponding angles are congruent.</i> MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar. <i>Clarification 1: Instruction includes showing that the corresponding sides are proportional, and the corresponding angles are congruent.</i>	Lesson 5	- I can identify the angle of rotation of a transformation of a figure on a plane. - I can identify the center of rotation of a transformation of a figure on a plane. - I can identify the direction of a rotation of a transformation of a figure on a plane. - I can distinguish a rotation from other transformations.
	Lesson 6	I can describe a single transformation that was applied.
	Lesson 7	- I can recognize the relationship between corresponding sides and corresponding angles of congruent and similar figures. - I can write congruence statements to relate corresponding parts following a rigid transformation. - I can write similarity statements to relate corresponding parts following a dilation.
	Lesson H1	- I can determine symmetries of reflection. - I can determine symmetries of rotation. - I can determine symmetries of translation.

Section 3: Constructions (Lessons 8-12)

Benchmarks Addressed	Learning Targets	
MA.912.GR.5.1 Construct a copy of a segment or an angle. <i>Clarification 1: Instruction includes using compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.</i> MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment. <i>Clarification 1: Instruction includes using compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.</i>	Lesson 8	- I can construct a copy of a line segment. - I can use the segment addition postulate.
	Lesson 9	- I can construct a copy of an angle. - I can use the angle addition postulate.
	Lesson 10	- I can construct the bisector of a segment. - I use the definition of a midpoint to solve problems.
	Lesson 11	- I can construct a perpendicular bisector. - I can use rigid transformations to show that any triangle formed by drawing two segments from the endpoints of a segment to a point on its perpendicular bisector is isosceles. - I can use the properties of a perpendicular bisector to solve problems.
	Lesson 12	I can construct an angle bisector.



Vertical Alignment

Grade 8 Unit 9
Transformations



Geometry 1 Unit 1
Basics of Geometry and
Constructions

Prior Course(s)	Course Level
MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.	MA.912.GR.2.2 Identify transformations that do or do not preserve distance. MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure. MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent. MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar. MA.912.GR.2.4(Honors only) Determine symmetries of reflection, symmetries of rotation and symmetries of translation of a geometric figure. MA.912.GR.5.1 Construct a copy of a segment or an angle. MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.

1.1 Defined and Undefined Terms – Part 1

Lesson Introduction				
 Benchmark	MA.912.GR.2.2 Identify transformations that do or do not preserve distance.		 Learning Targets	- I can use rigid motions to define parallel lines and perpendicular lines. - I can experiment with rigid motions on a plane. - I can use rigid motions to define a right angle.
 Benchmark Coherence	Building On		Working Toward	
	8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.		N/A	
 Essential Questions	- What is a line? - How do you label a line and a line segment? - What is a plane? - What are parallel lines? - What are perpendicular lines?			
 Lesson Narrative	This is the first part of a two-part lesson. Students use patty paper to explore facts about lines and planes, developing an essential foundation of motions on a plane that will be used throughout the course of Geometry. Students will engage in collaboration about how motions on a plane can be used to formulate definitions by observing key details of transformations. The lesson applies transformations to help students explore important terms and relationships that are foundational.			
 Component Summary	1.1.1 Warm-Up (~5 min.)	Estimate location of one billion on a number line (<i>SE p. 3</i>)	1.1.3 Collaboration (15 min.)	Define key foundational words of Geometry through motions on a plane (<i>SE p. 7</i>)
	1.1.2 Guided Instruction (25 min.)	Define key foundational words of Geometry through motions on a plane (<i>SE p. 4</i>)	1.1.4 Wrap-Up (~5 min.)	Reflection rating of lesson understanding (<i>SE p. 9</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Line - Line Segment - Plane - Parallel Lines - Right Angle - Perpendicular Lines - Skew Lines <u>Additional Terms:</u> - Reflection		- Patty Paper (Optional) Protractors (Optional) Card Stock (Optional) Sticky Notes	
			- Consider reviewing Jo Boaler’s research on how mistakes grow the brain. - Complete 1.1.2 Guided Instruction question 3 before class, saving each step on a separate piece of patty paper so you can quickly display it if needed. - Honors Geometry students may be able to complete lessons 1 and 2 together. - Create sentence stems that students can use to discuss strategies in 1.1.1 Warm-Up.	

 Teacher Tips	Common Misconceptions	Things to Consider
	• Students may struggle with the difference between a line and a plane. • Students may have a difficult time understanding that a line continues forever in both directions. • Some may have a difficult time visualizing a plane.	• Before starting 1.1.1 Warm-Up, consider putting a blank number line on a wall, possibly with chart paper or by projecting it. After students have completed the 1.1.1 Warm-Up, have a few students place sticky notes where they believe the number one billion is. Students may not be familiar with each other (or with you). Consider utilizing the 1.1.1 Warm-Up as an opportunity to establish, discuss, and practice classroom norms. • This lesson could easily be divided up in different ways so that while establishing class norms for your class students are also putting these norms into practice. • Creating a Word Wall may be helpful at the beginning of this unit. This will set a positive tone for creating Word Walls throughout the course.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time Starting with the first lesson of the course, students have many opportunities to work with a partner to help define terms and ideas. This Literacy Routine allows students to refine and clarify thoughts they have, allowing them to end the lesson with a solid grasp on the lesson content.	MA.K12.MTR.4.1: Discussion and Reflection Students will be working with partners and other pairs to develop and fine-tune definitions of key words. In 1.1.2 Guided Instruction, students write definitions, then are prompted to collaborate with classmates to fine-tune and verify their definition.
 Differentiate Instruction	For 1.1.2 Guided Instruction and 1.1.3 Collaboration, consider providing the copied images and partial steps on sheets of patty paper for those students who struggle with the physical dexterity needed to trace and fold the papers. This step will allow the students to make observations of the motions without hindrance.	
Lesson Notes:		

1.1.1 Warm-Up: A Billion!

Component Overview: Students estimate the location of 1 billion on a given number line and describe the strategy they used to determine the location.



1.1.1 Warm-Up: A Billion!

One billion is a big number. Starting the school year, it may feel like there are a billion seconds left in the year. When you started school there were 1.555×10^7 , or 15,550,000 seconds left in the school year.

A billion seconds is approximately 32 years!

1000000000	=	31.7097919838
Second		Calendar year

So, although it may feel like a billion, it really isn't.

1. What is the location of 1 billion on the line segment?



2. Describe your strategy for estimating the location of 1 billion on the line segment.

Answers will vary. I approached this question thinking that a billion was very close to the trillion mark. After thinking about the number of zeros, a trillion has three more zeros, making it a thousand times bigger than one billion. So my mark represents one thousandth of the way from zero to one trillion.



The movie “Paper Clips” deals with this very idea of relating large numbers to students. Showing or discussing this film may be a great way to connect with Social Studies or English and help students begin to develop the idea of how large numbers really are. Specifically, the movie is about a project that tried to illustrate to students the magnitude of the 6 million Jews killed during the Holocaust.



Consider using 1.1.1 Warm-Up to establish class norms on working with partners by pairing students to discuss their strategies. Establish the concept of mistakes as learning opportunities by using positive statements about the mistakes and how mistakes are an opportunity to learn.



If debriefing this activity prior to moving to 1.1.2 Guided Instruction, consider these questions:

- What value would be located in the middle of this line segment?
- How could this line segment be divided into smaller intervals?
- How many times smaller is 1 billion than 1 trillion?

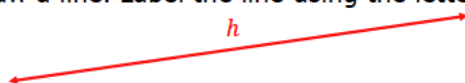
1.1.2 Guided Instruction: Definitions

Component Overview: After writing their best definition of a line segment and line, students are presented with facts about lines and planes. Using patty paper, students explore how motions on a plane are used to develop definitions.



1.1.2 Guided Instruction: Definitions

1. In the Warm-Up, the diagram is a line segment.
 - a. Discuss with your partner why it wasn't called a line.
Answers will vary. If a movie segment is a small "part" of a movie, then a line segment must be just a small part of a line.
 - b. Write your best definition of a line segment.
Answers will vary. A line segment is part of a line that is clearly beginning and ending at exact points.
 - c. Draw a line. Label the line using the letter h .



- d. Write your best definition of a line.
Answers will vary. A line is a way of connecting a bunch of points that continue in both directions for ever.
- e. Ask another partner pair the definitions they wrote. Record their definition and write your summary of their reason for the definition. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
Answers will vary.

Definition	My Summary of Their Reason	Initials
line segment		
line		



Question 1a is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address. 1.1.2 Guided Instruction is scaffolded to allow students to engage with content with their partner, the class, and with the teacher. Establishing an atmosphere where students may revise their thinking throughout instruction as they engage and discuss provides a safe atmosphere for students to explore and to be wrong.



Students are being asked to write their best definition of a line and line segment. The intent here is not to provide the definition for them and teachers should avoid any clarification of their definitions. In Lesson 2 of Unit 1, the "undefined terms" of Geometry will be introduced — point, line, and plane.

1.1.2 Guided Instruction: Definitions (continued)



Facts About Lines

- A line has infinite length, with no height and no width.
- A line is assumed to be straight.
- A line is considered one-dimensional.
- A line is considered an undefined term.
- A line goes on forever in both directions.



While discussing each of these facts about lines, avoid spending too much time on bullet 4, as an “undefined term” will be clarified in the next lesson. Focus more on the other four bullets and what they mean to the students.

A line has infinite length, thus, any line drawn should have arrowheads on both ends.

A line can be named by a single lowercase script letter, or by any two points which lie on the line.

2. Consider the two lines shown.



- Use the script lowercase letter m to name the purple line on the left.
line m
- Use the letters H and R to name the orange line on the right.
 $\overleftrightarrow{HR}$



Consider these discussion prompts after question 2:

- What does “infinite length” mean?
- How many points are on a line?



Facts About Planes

- A plane has infinite length and width, with no height.
- A plane is considered two-dimensional.
- A plane is considered an undefined term.
- A plane goes on forever.

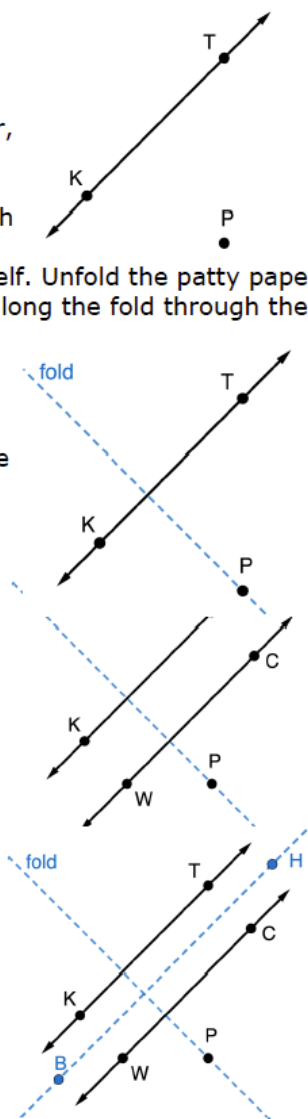
1.1.2 Guided Instruction: Definitions (continued)

A plane is named by a single uppercase letter written in script or by three points that do not lie on the same line.

3. Consider the line and point shown.

a. Complete the following:

- On a piece of patty paper, copy line $\overleftrightarrow{KT}$ and point P .
- Fold the patty paper so that the fold goes through point P and each part of line $\overleftrightarrow{KT}$ coincides with itself. Unfold the patty paper and draw a dashed line along the fold through the point, as shown below.
- Fold the dotted line on itself so that point P coincides with $\overleftrightarrow{KT}$.
- The resulting fold is a line that is parallel to the $\overleftrightarrow{KT}$. Use a writing utensil and straightedge to draw the parallel line. Label the line using letters W and C as shown below.
- Show the lines are parallel by folding the patty paper so that $\overleftrightarrow{KT}$ and $\overleftrightarrow{WC}$ coincide. Unfold the paper and draw a dashed line along the fold and name it $\overleftrightarrow{BH}$.



It may be important to read the directions here, replacing the word “coincides” with the words “lines up” or “match” to help students remember what the word “coincides” means.

1.1.2 Guided Instruction: Definitions (continued)

- b. Complete the statement.

$\overleftrightarrow{WC}$ is a
☐ rotation
☒ reflection
 of $\overleftrightarrow{KT}$.



The focus of this lesson is not on transformations. Avoid spending more than a few moments reviewing rotations and reflections, as they will be further explored in future lessons.

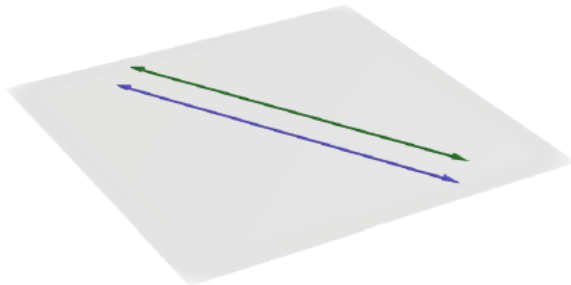
- c. Work with your partner to describe how the lines you drew on patty paper, $\overleftrightarrow{KT}$ and $\overleftrightarrow{WC}$, lie on the same plane.

Answers will vary. The patty paper represents a plane. Since I can draw $\overleftrightarrow{KT}$ and $\overleftrightarrow{WC}$ on the paper, then they are on the same plane.



Parallel lines do not intersect and lie in the same plane.

The diagram shows two parallel lines that lie on a plane. The plane is represented by the gray rectangle.



1.1.3 Collaboration: Definitions

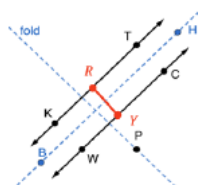
Component Overview: Students continue using the same patty paper to explore the definition of perpendicular lines.



1.1.3 Collaboration: Definitions

1. Continue using the patty paper from the previous component.

- a. Label a point on $\overleftrightarrow{KT}$ as R . Then, fold the patty paper along $\overleftrightarrow{BH}$ and copy point R to $\overleftrightarrow{WC}$ naming it Y . Unfold the paper and draw a line segment that connects points R and Y .



Example shown.

- b. What type of angle does $\overline{RY}$ form with the dashed line?

The line crosses at a right angle.

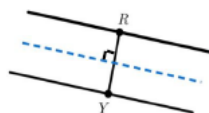


Students may not have ever used patty paper before. Allow students the space to make mistakes and to start over. As you circulate be prepared to help students who are struggling with using the patty paper.



A **right angle** is an angle measuring exactly 90° .

- c. Use a ruler to measure the distance between point R and the dashed line, and between point Y and the dashed line. Write the measurements, with units, on the diagram.



Answers will vary depending on the units. The distances between point R and the dashed line and point Y and the dashed line are the same, 1 cm each.

- d. Discuss with your partner: when you reflect an object across a line, will corresponding points be the same distance from the line of reflection? Summarize your discussion.

Answers will vary. Yes, a true reflection means that all parts are the same distance, so the distance something is from the line of reflection must also be the same for the reflected item.



Provide protractors for students to verify that the angle measures 90° . Or suggest to students to use a corner of a sheet of paper to verify the angle is a right angle.

1.1.3 Collaboration: Definitions (continued)

2. Complete the statements.

The patty paper is similar to a ☐ plane.
☐ reflection.

By reflecting a line across a line of ☐ reflection,
☐ rotation,

a parallel line was created. The parallel line ☐ is
☐ is not

the same distance from the line of ☐ reflection.
☐ rotation.

3. How is $\overline{RY}$ related to the dashed line?

$\overline{RY}$ is perpendicular to the dashed line because it is crossing the dashed line at exactly 90° .



After questions 1c and 1d, consider asking if this constant distance between parallel lines is important in the real world. Possibly in architecture, railroads, or other applications.



Perpendicular lines are lines that intersect at right angles.

The table shows multiple examples of perpendicular lines.

Description	Example of Perpendicular Lines
Perpendicular lines $\overline{KT}$ and $\overline{RB}$ shown in plane X .	
Perpendicular lines $\overline{KT}$ and $\overline{RB}$ shown in planes M and W .	
Perpendicular lines $\overline{KT}$ and $\overline{RB}$ shown in planes M , W and X .	

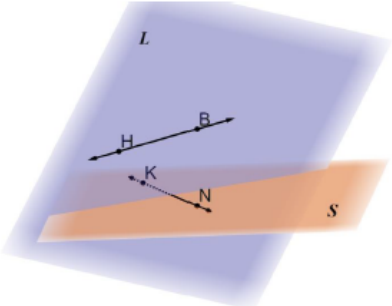
1.1.3 Collaboration: Definitions (continued)

4. If two lines are perpendicular, then any two adjacent angles formed are right angles.



Skew lines are lines that do not intersect and are not parallel.

The table shows an example of skew lines.

Description	Example of Skew Lines
Skew lines $\overleftrightarrow{HB}$ and $\overleftrightarrow{KN}$ shown in planes L and S .	



Consider asking students what they notice about the intersection of two planes. Or consider asking them to complete the statements:

Two lines in a plane intersect at _____. (a point)
The intersection of two planes is a _____. (a line)



To further help students visualize planes use a sheet of paper to represent a plane and two sheets of paper, where one has a slit, to show the two planes. Card stock or cardboard may also be helpful.

1.1.4 Wrap-Up: Reflection

Component Overview: Students self-reflect on the lesson and provide numerical values to rate their understanding.



1.1.4 Wrap-Up: Reflection

1. Rate yourself based on what you understood in today's lesson.

Answers will vary.

Skill	1 – I am just starting. 2 – I have some skills. 3 – I am almost there. 4 – I have it!
I can define parallel lines.	
I can define perpendicular lines.	
I can define a right angle.	

2. Rate yourself based on how you feel about the following statements.

Answers will vary.

Statement	1 – I struggle with this. 2 – I am ready to grow in this area. 3 – I am almost there. 4 – That's me!
I am willing to try new things.	
I persevere when something is challenging.	



For students who respond with many threes or fours, prompt them to discuss applications for each of the key words found throughout the lesson. Is there a real-world application, or a mathematical application? Having students "predict" test questions would also be effective.



Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.

1.2 Defined and Undefined Terms – Part 2

Lesson Introduction

 Benchmark	MA.912.GR.2.2 Identify transformations that do or do not preserve distance.		 Learning Targets	- I can understand the three undefined terms of geometry. - I can use rigid motions to define a straight angle. - I can experiment with rigid motions on a plane.	
 Benchmark Coherence	Building On			Working Toward	
	MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.			N/A	
 Essential Questions	- What is a definition? - What is meant by the phrase “undefined term”? - What are the properties of the three undefined terms of Geometry: point, line, and plane?				
 Lesson Narrative	This is the second lesson of a two-part lesson. Students will use patty paper to explore the rotation of a line segment and determine that a straight line has a degree measure of 180°. The undefined terms of Geometry are introduced along with their properties.				
 Component Summary	1.2.1 Warm-Up (~5 min.)	Justify the choice of two amounts (<i>SE p. 10</i>)	1.2.3 Bring It Together (10 min.)	Identify the three undefined terms of Geometry (<i>SE p. 11</i>)	
	1.2.2 Exploration (25 min.)	Explore the rotation of a line segment (<i>SE p. 10</i>)	1.2.4 Wrap-Up (~5 min.)	Students rate their level of confidence (<i>SE p. 12</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Straight Angle - Definition <u>Additional Terms:</u> - Undefined Term - Point - Line - Plane		- Patty Paper (Optional) Sticky Notes or Notecards		Consider using point, line, and plane cards or white boards as response tools for debriefing 1.2.3 Bring It Together question 2.

 Teacher Tips	Common Misconceptions • Students may not understand why a term is undefined. • Students may confuse the properties of a point, line, and plane.	Things to Consider • Students may struggle with the directions in 1.2.2 Exploration. Consider having samples that can be used for demonstration if needed. • Consider having students tape or paste their patty paper into their workbook.
	Literacy and Language Routines Poll the Class Consider displaying the prompts found in the table from 1.2.3 Bring It Together one at a time. Have students indicate their choice of answer either by holding up their answer on paper, a whiteboard, or using technology devices. Display the results of initial reports before students are asked to defend their choice.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.1.1: Participate in Effortful Learning In 1.2.2 Exploration, teachers should encourage students to participate actively in effortful learning to persevere when they are unsure of steps when using patty paper to construct different angles. Students should stay engaged and maintain a positive mindset while working on 1.2.2 Exploration. Teachers can help students support each other by providing sentence stems.
	 Differentiate Instruction	
Lesson Notes:		

1.2.1 Warm-Up: Would You Rather?

Component Overview: Students choose and justify their reasoning on whether they would rather have a stack of pennies or a stack of quarters to match their weight or height.



1.2.1 Warm-Up: Would You Rather?

1. Would you rather have pennies to match your weight or quarters to match your height?
Answers will vary.
2. Explain your reasoning.
Answers will vary.



Warm-Ups often provide all students with an entry-point and a safe place to be successful. Allow students to answer without the fear of being wrong. Consider providing the US Mint's coin specifications for students who want more details. Consider using 1.2.1 Warm-Up to further establish class norms and routines.

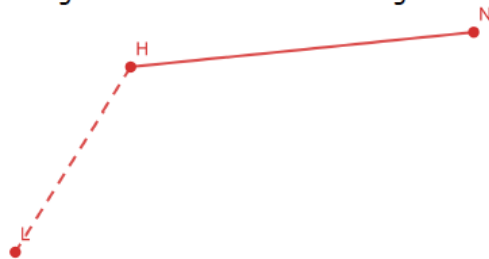
1.2.2 Exploration: Definitions

Component Overview: Students will explore rotating a line segment using patty paper.

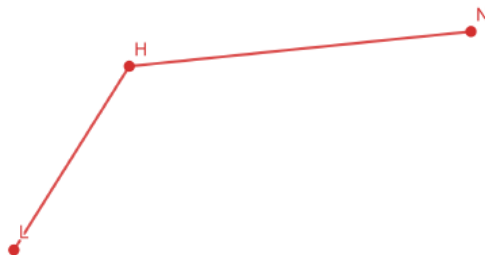


1.2.2 Exploration: Definitions

- Use two sheets of patty paper to complete the following.
 - Draw $\overline{HN}$ on one piece of patty paper.
 - Draw $\overline{KL}$ on the other piece of patty paper.
 - Lay the piece of patty paper with $\overline{HN}$ drawn on it on top of the piece with $\overline{KL}$ drawn on it. Place the papers so that the points H and B are on top of each other and the line segments coincide.
 - Place your writing utensil on endpoint H .
 - Rotate the top sheet of patty paper until the two line segments make an obtuse angle.



- Draw the obtuse angle you created below.



- When you rotated the top sheet, did the line segment change in size?

No



Consider beginning the lesson with a short overview of things that students remember from Lesson 1. Allow students to use their workbook to share ideas. Consider using a Think-Pair-Share routine by prompting students to write down 3 things they learned from Lesson 1. Have students share their thinking in a low-stakes way with one partner. Then each pair can share one thing with the class. As you circulate eavesdrop on the partner conversations so that you can purposefully select students to share with the class. This review will be helpful for students during question 2 of 1.2.3 Bring It Together.



To support the opportunity for students to expand their vocabulary, ask the following questions:

- When is the angle acute?
- At what position is a right angle formed?
- Where is the angle obtuse?

1.2.2 Exploration: Definitions (continued)

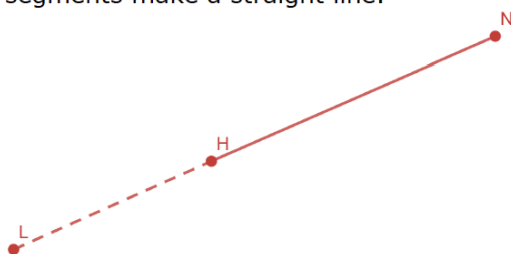
4. When you rotated the top sheet, did the line segment change in position?

Yes

5. Which part(s) of the obtuse angle has two points?

Each side of the obtuse angle, segment HN and segment HL , have two points.

6. Rotate the top sheet of patty paper until the two line segments make a straight line.



7. Name the resulting line segment using the endpoints.

The resulting line segment is $\overline{NL}$.

8. Work with your partner to estimate the number of degrees you rotated the top sheet of patty paper from its original position on top of line segment BL .

Answers will vary. I estimate that I rotated the shape about 180° .



A **straight angle** is an angle measuring exactly 180° .

9. Based on the rotation and the definition of a straight angle, what "fact" about a line can be added to the "Facts about Lines" in the previous lesson?

A line measures 180° .



Students are experiencing transformations that preserve distance. This concept of preserving distance is not formally established but is used to build student understanding of definitions.

1.2.3 Bring It Together: Undefined Terms

Component Overview: Students are introduced to what a definition is and why there are undefined terms in Geometry. Then, the three undefined terms are matched to their properties.



1.2.3 Bring It Together: Undefined Terms



A **definition** is an exact statement, written formally, of the meaning or description of a word.

There are three terms in geometry that are considered “undefined terms.”

A term is considered undefined if, when explaining the term, the explanation relies on the term itself or a close synonym. This is sometimes called a circular definition.

1. In the previous lesson you wrote a definition for the term “line.” Review your definition and determine if your definition is considered circular.

Answers will vary. I would consider my definition circular.

Although we have “facts about lines,” “line” is an undefined term in geometry. The other two undefined terms in geometry are “point” and “plane.” Look back through the previous lesson to notice how often these terms were used without providing a definition.



In Geometry, line is considered an undefined term. Other courses may have provided a definition based on the properties of a line but those are not considered to be accepted definitions as part of this course.

Consider having students write their definitions on sticky notes or notecards for you to use as examples to show why the definitions are circular. Question 1 is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.

1.2.3 Bring It Together: Undefined Terms (continued)

2. Work with your partner to label each description as a line, plane, or point.

Description	Undefined Term
has two dimensions	☐ Line ☐ Point ☒ Plane
has no length, no width, and no height	☐ Line ☒ Point ☐ Plane
is often drawn as a four-sided figure	☐ Line ☐ Point ☒ Plane
has infinite length, no width, no height	☒ Line ☐ Point ☐ Plane
named by an ordered pair in the coordinate plane	☐ Line ☒ Point ☐ Plane
indicates a location or position	☐ Line ☒ Point ☐ Plane
has one dimension	☒ Line ☐ Point ☐ Plane
has infinite length, infinite width, and zero height	☐ Line ☐ Point ☒ Plane
has no dimension or actual size	☐ Line ☒ Point ☐ Plane



Use the literacy routine “Poll the Class” to gauge student understanding of the three undefined terms of Geometry. Display the prompts found in the table one at a time and have students indicate their choice of answer either by holding up their answer on paper or a whiteboard or using technology devices. Display the results of initial reports before students are asked to defend their choice.

1.2.4 Wrap-Up: Reflection

Component Overview: Students reflect on their confidence with the new material by providing a rating of 1 – 4.



1.2.4 Wrap-Up: Reflection

1. Rate yourself based on what you understood in today's lesson.

Answers will vary.

Skill	1 – I am just starting. 2 – I have some skills. 3 – I am almost there. 4 – I have it!
I can understand the three undefined terms of geometry.	
I can define a straight angle.	

2. Rate yourself based on how you feel about the following statements.

Answers will vary.

Statement	1 – I struggle with this. 2 – I am ready to grow in this area. 3 – I am almost there. 4 – That's me!
I am a student learner.	
I am a mathematician.	



Cruising Clipboard

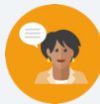
Walk around the room and quickly make note of the sum of the values for the first box.

- When most students are having sums of ten or higher, students are confident.
- If students are closer to a six to nine range, they may benefit from another opportunity to solidify these definitions.
- Any time scores are five or under, consider reteaching this lesson, or intentionally integrating these definitions early in the next few lessons.

1.3 Isometry

Lesson Introduction

 Benchmark	MA.912.GR.2.2 Identify transformations that do or do not preserve distance.		 Learning Targets	- I can define rigid motions. - I can show rigid motions by mapping using arrows on the plane. - I can explain why isometry preserves distance.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.		N/A	
 Essential Questions	- When a shape is moved, does it change size? - What transformations result in a shape the remains the same size? - Can I identify the specific transformations that map a preimage to the image?			
 Lesson Narrative	This lesson allows students to engage with rigid transformations in a very kinesthetic and intentional way. Key vocabulary is defined that will be critical for learning throughout the remainder of the course. The primary learning outcome of this lesson is to identify transformations that do and do not preserve distance.			
 Component Summary	1.3.1 Warm-Up (~5 min.)	Notice and wonder of an image moving on pages (<i>SE p. 13</i>)	1.3.3 Bring It Together (10 min.)	Defining words associated with rigid motion (<i>SE p. 17</i>)
	1.3.2 Exploration (20 min.)	Exploring translations, reflections, and rotations with patty paper (<i>SE p. 14</i>)	1.3.4 Collaboration (10 min.)	Mapping rigid motion on gridded paper (<i>SE p. 18</i>)
	1.3.5 Wrap-Up (~5 min.)	Reflection on the lesson (<i>SE p. 19</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Reflection - Rotation - Translation - Isometry - Rigid Transformation - Rigid Motion - Congruent - Mapping - Line of Reflection Additional Terms: N/A		- Flipbook Video - Patty Paper - Ruler or Protractor (Optional) Sticky Notes	
			Additional Preparation N/A	

 Teacher Tips	Common Misconceptions • Students may believe that if a shape looks the same, it is the same and measurement is not required to check. • Students may think diagonal movement is the same as a translation.	Things to Consider • For the transformations in 1.3.2 Exploration, consider asking students to label the vertices of each of the 5 images. • Consider a Foldable, Frayer Model, or type of interactive notebook for definitions throughout. • Consider allowing students to work through 1.3.2 Exploration and 1.3.4 Collaboration without worrying about having the “right” answers. By circulating and making notes about common errors and student misconceptions you can discuss these common errors and misconceptions in future lessons. Unit 1 is designed to help student explore some of the fundamental ideas of Geometry by utilizing transformations to have students experience them. The content repeatedly spirals back through the concepts.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder For 1.3.1 Warm-Up, we use a Notice and Wonder to offer low-stakes questions inviting students to engage in the idea of rigid transformations. The animation presented is intended to create curiosity that could motivate more students to connect with the material in ways they previously would not have.	MA.K12.MTR.4.1: Discussion and Reflection Students will be implementing the Discussion and Reflection standard in 1.3.2 Exploration and 1.3.4 Collaboration. Students engage in discussions about mathematical thinking. Help to establish a culture where students ask questions to their peers and to you. Consider opportunities to present student work to help deepen student understanding.
 Differentiate Instruction	While developing a plan for helping students comprehend each of the six key words in 1.3.3 Bring It Together, consider having students develop a folded Frayer Model. This allows students to have a kinesthetic connection with these key words as they engage with each definition. Once completed, creating a pictorial representation of each word will also offer a visual entry-point for students.	
Lesson Notes:		

1.3.1 Warm-Up: Animation Flipbook

Component Overview: Play the Flipbook video for the entire class. After students watch the video (0:42 minutes), prompt them to reflect on the profession by answering the two questions independently.



1.3.1 Warm-Up: Animation Flipbook

An animation flipbook is a small book of drawings with a series of pictures that appear to move when the pages are flipped in rapid succession.



The video shows a series of pictures of two clouds. A few screenshots are shown.



1. What do you notice about the two clouds?
Answers will vary. The cloud on the right is happy, and the cloud on the left is sad.
2. At one point the two clouds seemed to move up and down. How do you think the artist was able to do that?
Answers will vary. The artist made small changes on each page to make it look like the clouds are moving when flipping through the book.



1.3.1 Warm-Up is designed to simply engage students. If students are missing the point or becoming confused, do not spend extra time clarifying here. This should be a quick, fun activity for students to become excited about the content. The video shows that the figures move but their size and shape do not change. Making this connection in 1.3.3 Bring It Together may be helpful to students. A further connection would be the application to video games where characters move but their size/shape do not change.



Should students become excited about this idea of a flipbook, it can make a great homework activity if it does not take much time away from this lesson. Consider prompting students to develop a flipbook that uses a key shape they learned from a previous class.

1.3.2 Exploration: Isometry

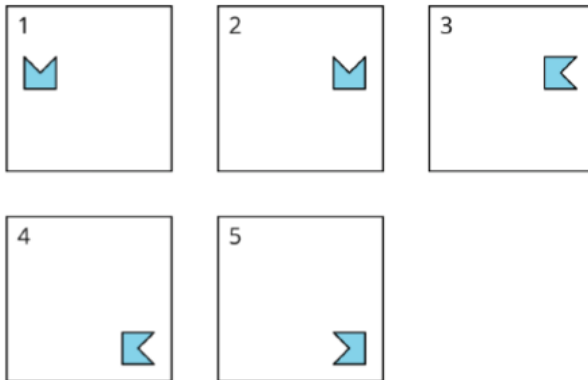
Component Overview: Students will explore moving the position of the image of a polygon using patty paper to determine if size and shape stays the same or changes for each type of movement.



1.3.2 Exploration: Isometry

Work with your partner to complete the following.

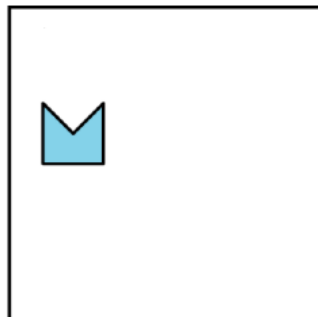
- Five frames that feature a polygon in different positions are shown.



Describe how the polygon moves from one frame to the next.

Movement from...	Description
Frame 1 to Frame 2	<i>Slide to the right</i>
Frame 2 to Frame 3	<i>Turn to the right</i>
Frame 3 to Frame 4	<i>Slide down</i>
Frame 4 to Frame 5	<i>Turn the other way</i>

- The diagram shows the position of the polygon from Frame 1.



This exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the 1.3.3 Bring It Together. Students may revise their thinking throughout the exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group 1.3.3 Bring It Together.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

Consider asking student to label corresponding vertices in each figure or color each corresponding edge the same color. Look for:

- Does the student translate or reflect from step 1 to step 2?
- Does the student translate or reflect from step 3 to step 4?
- Does the student reflect or rotate from step 4 to step 5?

1.3.2 Exploration: Isometry (continued)

- a. Label two pieces of patty paper as **A** and **B**. Then, complete the steps for each piece of patty paper.

Patty paper A

- Copy the polygon.
- Move patty paper **A** to display the polygon as it is shown in Frame 2.

Patty paper B

- Copy the polygon and the square that surrounds it.
- Fold the patty paper vertically so that the top corners of the square match up.
- With the patty paper folded, draw the copy of the polygon onto the other half of the patty paper.

- b. Complete the statements.

To draw the polygon as it is shown in Frame 2,

patty paper **A** was moved

- ☐ up.
- ☐ left.
- ☒ right.
- ☐ down.

When patty paper **B** was folded, the polygon could be traced so that the new polygon is as shown in Frame 2.

The original polygon was

- ☒ flipped across
- ☐ rotated around

the fold of the paper.

The fold of the paper was the

- ☐ center of rotation.
- ☒ line of reflection.



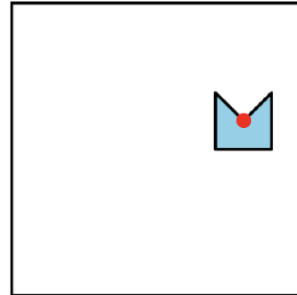
Students may wonder if the border of the square should be traced. It is not necessary for the border to be traced but a student may feel more comfortable doing so.



Scaffolding student responses help students to explain concepts they are just learning. Encourage students to read their completed sentence out loud to their partner. This will help students hear their own response and allow their partner to consider if they agree with the answer.

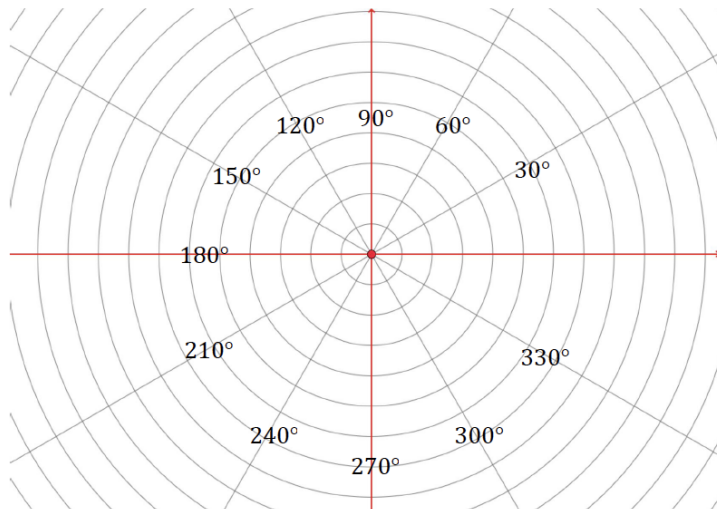
1.3.2 Exploration: Isometry (continued)

3. The diagram shows where the polygon is in Frame 2.



a. Complete the steps to find the point of rotation.

- Label a piece of patty paper as **C**.
- Copy the polygon onto patty paper **C**.
- Use the point of your writing utensil to find the point of rotation so the polygon can be rotated to look like Frame 3.
- Place a dot on the diagram above to show the point of rotation.
- Place patty paper **C** on the circular graph paper so the point of rotation is where the two red axes meet, and the polygon is oriented as in Frame 2.
- Rotate patty paper **C** so that it looks like Frame 3.



Students may benefit from a demonstration of how to use the point of their writing utensil to spin the patty paper around to find the point of rotation.

1.3.2 Exploration: Isometry (continued)

- b. Complete the statements.

To move the polygon so it looks like Frame 3, rotate the diagram on patty paper $\text{C } 90^\circ$ degrees clockwise $\cup$ or 270° degrees counterclockwise $\cap$.

The center of rotation is

- ☐ the top left vertex.
- ☐ the top right vertex.
- ☐ the bottom left vertex.
- ☐ the bottom right vertex.
- ☒ the center of the polygon.

4. Use patty paper, a ruler, or a protractor as necessary to complete the following.

- a. Complete the table by determining if the polygon changed in size or shape as it moved.

Movement from...	Did the size change?	Did the shape change?
Frame 1 to Frame 2	No	No
Frame 2 to Frame 3	No	No
Frame 3 to Frame 4	No	No
Frame 4 to Frame 5	No	No

- b. What tools or information were used when determining if the size changed?
Ruler and Patty Paper
- c. What tools or information were used when determining if the shape changed?
Protractor and Patty Paper
- d. Discuss with your partner if the measurement of angles in the polygon changed from Frame 1 to Frame 5.
No, they did not change.



Listen for students who think they can just “look” to determine if the size or shape changed. Help students understand the importance of precision.

1.3.3 Bring It Together: Definitions

Component Overview: Students can summarize or synthesize their findings in this activity. Translations capture horizontal and vertical transformations as seen in questions 1 and 2 in 1.3.2 Collaboration, while examples of rotations can be found in questions 3-5 of 1.3.2 Collaboration.



1.3.3 Bring It Together: Definitions

The movements of the polygon in the five frames are known as reflection, rotation, and translation. During the lesson a reflection was referred to as a flip, which is an informal definition. Another informal definition used was "is moved," which is formally known as a translation.



A **reflection** is a transformation that produces the mirror image of a geometric figure across a line of reflection.



A **rotation** is a transformation of a figure by turning it about a center point or axis. The amount of rotation can be expressed in the number of degrees. The direction of the rotation for two-dimensional figures can be expressed as clockwise or counterclockwise.



A **translation** is a transformation in which every point in a figure is moved in the same direction and by the same distance.

Isometry is a transformation that preserves distance and angle measure.

1. Discuss with your partner the definition of isometry. Determine which transformations — reflection, rotation, or translation — preserve distance and angle measure. Summarize your discussion. *These three transformations preserve distance and angle measurements.*



A **rigid transformation** is a transformation of points in space consisting of a sequence of one or more translations, reflections, or rotations. Rigid transformations preserve distances and angle measures (congruency).



A rigid transformation is also called a **rigid motion**. Rigid means that something is not able to be changed. Isometry is composed of rigid motions.



Figures having exactly the same shape and size are **congruent**. Equivalently, two figures are congruent if one can be mapped to the other using a rigid transformation.



Consider a foldable, Frayer Model, or type of interactive notebook for these definitions. Examples, non-examples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

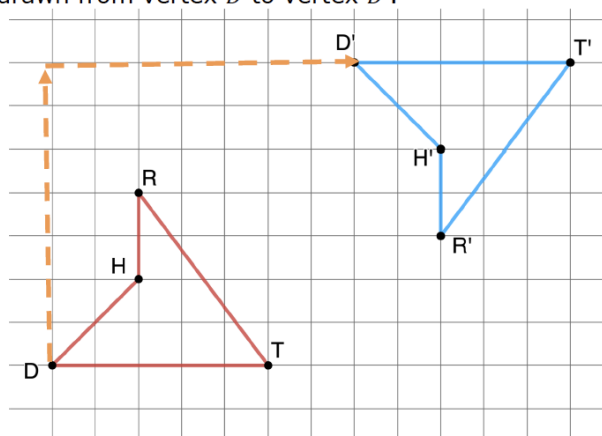
1.3.4 Collaboration: Isometry

Component Overview: Students map translations and rotations on gridded paper to determine that these transformations map all points on the figure.



1.3.4 Collaboration: Isometry

1. A diagram of $RTDH$ and $R'T'D'H'$ is shown. Arrows are drawn from vertex D to vertex D' .



$R'T'D'H'$ is read as "r prime, t prime, d prime, h prime". This notation is often used when a figure has been transformed.

The arrows show a possible path of movement. For the movement to be correct, the same path should exist for all points on a diagram.

Discuss with your partner if all of the points on $RTDH$ can be moved using the same path as point D .



A **mapping** is a standardized form of showing a transformation.



This is students' first introduction to mapping. Mapping shows the movement of a transformation. Mapping can be done with horizontal and vertical vectors or with a vector that is the hypotenuse of the triangle formed by the horizontal and vertical vectors. In preparation for students to describe a transformation algebraically using horizontal and vertical vectors may be easier for students. Algebraic descriptions using coordinate notation such as $(x, y) \rightarrow (x + h, y + k)$ are introduced in Unit 4. It is not necessary to use this notation in this unit but if your students are ready for this notation exposing them to the idea may lay a foundation for later units.

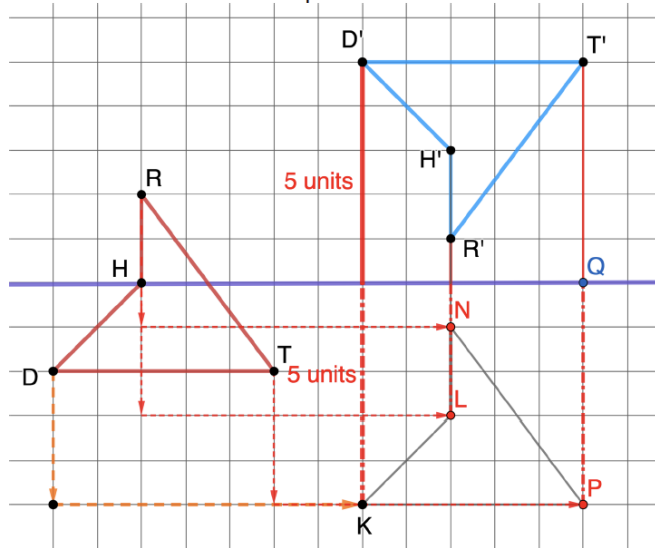
Also, note using notation, such as $(x, y) \rightarrow (x + h, y + k)$ can be called many things such as: algebraic descriptions, coordinate notation, or algebraic descriptions using coordinate notation. Utilize all of these names when discussing this representation.



This is a discussion opportunity. Students do not need to write an explanation here. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

1.3.4 Collaboration: Isometry (continued)

2. A diagram of $RTDH$ and $R'T'D'H'$ is shown. Arrows are drawn from vertex D to point K .



- Draw a line segment from point K to the purple line. What is the length of the line segment?
Shown on graph.
- Draw a line segment from point D' to the purple line. What is the length of the line segment?
Shown on graph.
- Describe the relationship each line segment has to the purple line.
Both line segments are perpendicular to the purple line.



If corresponding points have the same distance from a line then the line is the **"line of reflection."**



Students may make errors during 1.3.4 Collaboration, including mapping incorrectly. Allow the opportunity for them to correct their own thinking as they work through with their partner. Part of what students may realize by the end of the 1.3.4 Collaboration is that mapping one figure to another may be shown using the vertices but all of the points of the figure follow that same mapping. This idea is continually developed in Unit 1. Do not be afraid to let students make mistakes. As you circulate keep track of common errors by students that you can address in later lessons.



If students wonder it may be helpful to specify that the line segment should be the shortest distance from the point to the line, also known as the perpendicular distance.

1.3.4 Collaboration: Isometry (continued)

- d. For each of the other vertices on $RTDH$, draw a new point using the same movements done to map point D to point K . Then show the reflection to $R'T'D'H'$ using line segments.

Shown on graph.

- e. Discuss with your partner if the mapping of **all** of the points on $RTDH$ can be done using the same path and reflection, or if the mapping only works for the vertices. Summarize your discussion.

Answers will vary. Yes, all points can be mapped this way.



The focus of this component is to illustrate that “same size, same shape” is an inefficient definition of congruence. The foundation for the definition of congruence is conceptually taught early so that it can be applied throughout the course.

1.3.5 Wrap-Up: Reflection

Component Overview: Students respond to prompts of what they learned and what they still are wondering about from the lesson.

1.3.5 Wrap-Up: Reflection

1. Complete the table.
Answers will vary.

The terms and information used in the exploration that I knew were ...	Something I wondered during the exploration was ...	Something I learned during the exploration was ...



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons. Consider writing each student a brief note about their response or to have students anonymously share one of these reflections on a sticky note where you can write a positive note. Then, display all of the notes for students to view.

1.4 Dilations and Partitions

Lesson Introduction

 Benchmark Focus	MA.912.GR.2.2 Identify transformations that do or do not preserve distance.		 Learning Targets	- I can dilate a line segment. - I can define midpoint using dilation and understand partitions. - I can relate scale factor to dilation.
 Benchmark Coherence	Building On MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.		Working Toward N/A	
 Essential Questions	- How can knowing a scale factor between two figures help determine a segment length? - How is a scale factor related to a dilation? - How is dilation used to partition a segment?			
 Lesson Narrative	Students explore situations to determine how many smaller segments need to be copied to create a larger segment. Relating the principle of scale factor to dilations, students will partition segments and define midpoint.			
 Component Summary	1.4.1 Warm-Up (~10 min.)	Floyd Norman, Animator Career Profile (<i>SE p. 20</i>)	1.4.3 Guided Practice (15 min.)	Applying scale factor to determine length (<i>SE p. 24</i>)
	1.4.2 Exploration (15 min.)	Using dilation to partition a line segment (<i>SE p. 21</i>)	1.4.4 Bring It Together (5 min.)	Using scale factor to define midpoint (<i>SE p. 26</i>)
	1.4.5 Wrap-Up (~5 min.)	Reflection on whether dilation is an isometry (<i>SE p. 27</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Dilation - Partition - Scale Factor - Midpoint <u>Additional Terms:</u> - Isometry		- Career Profile Animator Floyd Norman - Patty Paper - Calculator	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may not remember the Pythagorean theorem.	- Consider learning more about Floyd Norman to share his accomplishments with your students. - Consider further discussing with your students how Floyd Norman shows a growth mindset. The lesson uses the idea that mistakes are opportunities to learn. Focus on how Floyd Norman tried things and was willing to fail in order to learn.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns Students will work through 1.4.2 Exploration with a partner. Students should take turns estimating and discussing the correct procedure. As students discuss, they will decide if the estimations or procedures are reasonable and follow the expected procedure for the assigned activity. Students should continue this method while working through 1.4.3 Guided Practice.	MA.K12.MTR.1.1: Participate in Effortful Learning Students will have opportunities throughout the lesson to cultivate a growth mindset. In 1.4.1 Warm-Up, students listen to an interview with Floyd Norman where he shares how he was able to learn from mistakes and overcome obstacles. In 1.4.2 Exploration students will recognize which student had an error in their thinking and then write an explanation of the error. Throughout the lesson students help and support each other while attempting a new method.
 Differentiate Instruction	For 1.4.3 Guided Practice, consider providing an alternate grid and changing the length of $\overline{MS}$ to 15 units. This will simplify the scale factor and keep that content focused on dilations and partitions, not fractions and decimals.	
Lesson Notes:		

1.4.1 Warm-Up: Floyd Norman

Component Overview: Play the Career Profile video on Disney Animator Floyd Norman for the entire class. After students watch the video (3:20 minutes), prompt them to reflect on the interview by answering the three questions independently.



1.4.1 Warm-Up: Floyd Norman

Floyd Norman was named a Disney Legend in 2007. In 1957, Norman was employed as an inbetweener on *Sleeping Beauty*. An inbetweener is an animator who draws the intermediate frames between two key frames. The inbetweener's drawings create the illusion of motion.

When asked, "What is the key to growing as an artist?" Norman said, "As a student, I think the most important thing is having a willingness to learn."

1. On a scale of 1 to 10, how willing are you to learn?

Answers will vary. I am a 10, ready to learn at any opportunity.

When asked about his career Norman stated, "Making mistakes is how we grow, how we learn." He said he had no confidence, but at least he made the first step.

2. Describe a time in your life when you had no confidence, but you made the first step and learned from making mistakes.

Answers will vary. I had no confidence when I tried out for the soccer team. I did try out anyway and learned what I needed to do to make the team the next year.

Norman described a time when Walt Disney asked him to write. Norman made these two statements as he reflected on doing something he did not think he could do.

- "The only obstacle you face is yourself."
- "If you are willing to do the work, make that commitment, then I don't believe there is an obstacle."

3. What subject(s) are obstacles for you?

Answers will vary. The subject that is hardest for me is Chemistry.



Floyd Norman is considered a Disney Legend. During the interview he shares how he wasn't always confident but he tried something anyway. Students may have fears about learning Geometry. During this unit help students build a culture of growth mindset.

1.4.2 Exploration: Dilating a Line Segment

Component Overview: Students will determine how many segments can be copied to make a segment of another length and how this is related to dilations. Using scale factors, students will partition lines to find lengths.



1.4.2 Exploration: Dilating a Line Segment

1. The diagram shows points W , W' , and W'' .

- W W' W''
- What do you notice about the points?
Answers will vary. They all are in a line and have the marks ' and ''.
 - Estimate the number of times the distance between W and W' will fit between W' and W'' .
Answers will vary. I estimate about three or four W to W' will fit between W' to W'' .
 - Two students described the relationship $\overline{WW''}$ has with $\overline{W'W''}$.

Jaquaze	Andrés
$\overline{WW''}$ is three times as long as $\overline{WW'}$.	$\overline{WW''}$ is four times as long as $\overline{WW'}$.

Discuss with your partner which student is correct and the mistake the other student made. Write a brief note to the student who made the mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary. Jacquaze, it seems you found that three segments the length of $\overline{WW'}$ would fit between W' and W'' . That means $\overline{WW''}$ is four times as long as $\overline{WW'}$ because the total distance is four times as long.



1.4.2 Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next components. Students may revise their thinking throughout the exploration.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group 1.4.3 Guided Practice or 1.4.4 Bring It Together.

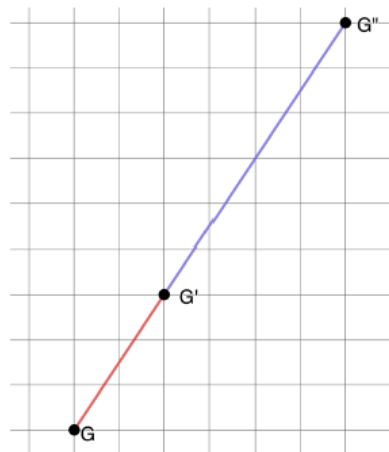
Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.



If students have a hard time understanding why it is four times and not three, ask the student to try placing a number like five in for the length of $\overline{WW'}$. "If the length of $\overline{WW'}$ is 5 units, then would the length of $\overline{WW''}$ be 15 or 20 units?"

1.4.2 Exploration: Dilating a Line Segment (continued)

2. The diagram shows points G , G' , and G'' .



- a. Use patty paper to determine the number of copies of $\overline{GG'}$ that will fit on $\overline{GG''}$.

Three copies of $\overline{GG'}$ will fit on $\overline{GG''}$.

- b. Discuss with your partner how to use the grid to determine the number of copies of $\overline{GG'}$ that will fit on $\overline{GG''}$. Summarize your discussion.

Answers will vary. $\overline{GG'}$ is three squares up and two to the right. If we follow the graph up and to the right repeatedly, we will find that three $\overline{GG'}$ will fit on $\overline{GG''}$.

- c. Ask a classmate how they would use the grid to determine the number of copies of $\overline{GG'}$ that will fit on $\overline{GG''}$. Record their description and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your description and summary are correct.

Answers will vary.



Using the Take Turns Literacy and Language Routine, students should work through question 2b by taking turns with their partner. Each time they are not talking, students should be verifying their partner's strategy by following their directions.

Description	My Summary of Their Reason	Initials

1.4.2 Exploration: Dilating a Line Segment (continued)



A **dilation** is a proportional increase or decrease in size in all directions.

- d. Use the definition of dilation to describe why $\overline{GG''}$ is a dilation of $\overline{GG'}$.

This is a dilation because $\overline{GG''}$ is exactly three times larger than $\overline{GG'}$.

- e. Fill in the blanks to complete the statements.

G' divides $\overline{GG''}$ into a ratio.

The ratio of $\frac{GG'}{GG''}$ is $\frac{1}{3}$.



Partition means to separate or to divide. A line segment can be partitioned into smaller segments. A ratio can be used to describe the partition.

- f. Use the Pythagorean theorem to complete the table.

Segment	Length
$\overline{GG'}$	$\sqrt{13}$
$\overline{G'G''}$	$2\sqrt{13}$
$\overline{GG''}$	$3\sqrt{13}$

- g. Point G' is $\frac{1}{3}$ of the distance from G to G'' .
- h. What value can be multiplied to the length of $\overline{GG'}$ to determine the length of $\overline{GG''}$?
3



The **scale factor** is the constant that is multiplied by the length of each side of a figure to produce an image that is the same shape as the original figure.



Students have had previous exposure in mathematics courses to the words, “dilation” and “scale factor,” but may have not been exposed to “partition.” Have the students connect their knowledge of the word partition in a non-mathematical sense prior to introducing this mathematical definition.



Students may provide $\sqrt{52}$ as the length of $\overline{G'G''}$ and $\sqrt{117}$ as the length of $\overline{GG''}$. These are equivalent to the answers shown on the key and show that students know how to correctly apply the Pythagorean theorem to find distance on the coordinate plane from prior courses. Students also learned to simplify radicals in Algebra 1. Based on observation and lesson timing, determine whether this is a skill that needs to be revisited in this lesson or in the future.

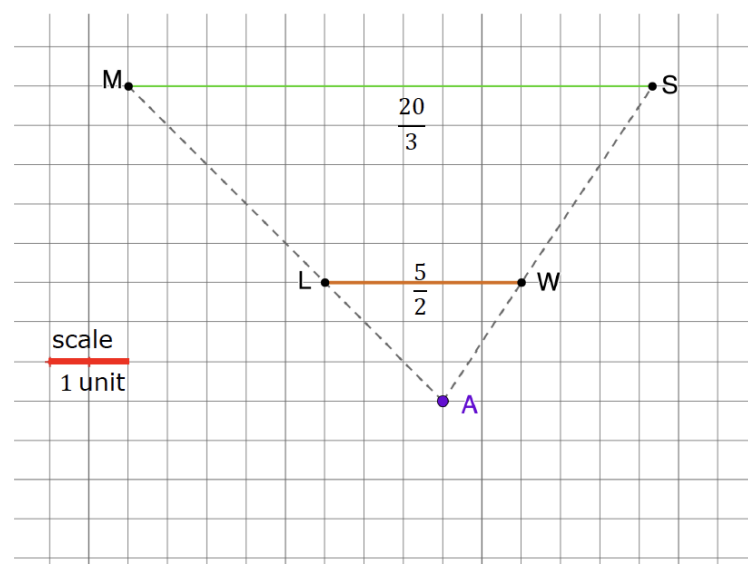
1.4.3 Guided Practice: Scale Factor

Component Overview: Students will determine the scale factor and determine all segment lengths of two overlapping triangles. They will work through questions about the length ratios to help develop the idea that scale factors are identical for all corresponding parts of similar figures.



1.4.3 Guided Practice: Scale Factor

$\overline{LW}$ is dilated to create $\overline{MS}$. The center of dilation is point A.



- $\overline{LW}$ has a length of $\frac{5}{2}$ units and $\overline{MS}$ has a length of $\frac{20}{3}$ units. What is the scale factor that was used to create $\overline{MS}$?
 $\frac{8}{3}$
- Use the Pythagorean theorem to complete the table. Round values to the nearest tenth of a unit.

Segment	Length	Segment	Length
$\overline{AL}$	2.1	$\overline{AW}$	1.8
$\overline{LM}$	3.5	$\overline{WS}$	3.0
$\overline{AM}$	5.6	$\overline{AS}$	4.8



Students may struggle with fractions. This problem allows teachers to see those potential issues to determine if remediation is needed for students to interact with the types of problems expected in the course. When determining the segment lengths in question 2, students can use the fractional values or convert them to decimals, if desired. Line segment WS is given so that students can use that given value to find AS by adding the value they find for AW to WS, if needed.



Considering the dilation of a line segment in this way by looking at the relationships between the distance between the endpoints and the center of dilation helps students focus on the multiple effects of dilations. Through this problem, students explore the relationship between distance and scale factor. Allow students to share how they are solving the problem, so they are exposed to a variety of ways of thinking. For example, some students may use slope while others use the Pythagorean theorem, which allows for discussion of the relationship between the two.

1.4.3 Guided Practice: Scale Factor (continued)

3. Point L is $\frac{3}{8}$ of the distance from A to M .
4. What value can be multiplied to the length of $\overline{AL}$ to determine the length of $\overline{AM}$?
 $\frac{8}{3}$
5. Point W is $\frac{3}{8}$ of the distance from A to S .
6. What value can be multiplied to the length of $\overline{AW}$ to determine the length of $\overline{AS}$?
 $\frac{8}{3}$
7. What do you notice about each of the following ratios?
 - $LW:MS$
 $\frac{3}{8}$ is the ratio.
 - $AL:AM$
 $\frac{3}{8}$ is the ratio.
 - $AW:AS$
 $\frac{3}{8}$ is the ratio.
8. Work with your partner to explain how to find the scale factor of a dilation.
Answers will vary. Find the ratios of each pair of corresponding side lengths. They should be equivalent. Determine the order of the ratio for the scale factor depending on which is the preimage and which is the image. For example, the scale factor from line segment LW to line segment MS is $\frac{8}{3}$, while the scale factor from MS to LW is $\frac{3}{8}$.



Students will likely confuse when to use $\frac{3}{8}$ and when to use $\frac{8}{3}$. When monitoring the students, if large numbers of students mix these up, bring the class back to whole-group discussion to clarify which is the scale factor and which is the ratio of the lengths.

1.4.4 Bring It Together: Determining Scale Factor

Component Overview: Students decide on their favorite strategy for finding scale factors. Students then dilate a line segment using a scale factor of two to relate that to the midpoint.



1.4.4 Bring It Together: Determining Scale Factor

As the class shares strategies for finding scale factor, choose a strategy other than yours to describe.

Answers will vary.

Strategy	I Liked the Strategy Because...



Ask different pairs of students to share their strategies. Options may include ratios of length, slope, or other methods they may develop.

1. $\overline{XR}$ and $\overline{RK}$ are shown.



a. What is the scale factor that can be used to dilate $\overline{XR}$ to create $\overline{XK}$?

2

b. What is the scale factor that can be used to dilate $\overline{XK}$ to create $\overline{XR}$?

$\frac{1}{2}$

c. Describe point R in relation to $\overline{XK}$.

Point R is in the middle of the line. It is halfway between points X and K .

The **midpoint** of a line segment is the point that divides a line segment into two equal segments.

1.4.5 Wrap-Up: Cool Down

Component Overview: Students describe whether a dilation is an isometry or not.



1.4.5 Wrap-Up: Cool Down

1. Isometry is a transformation that preserves distance and angle measure. Write a short note that describes whether a dilation preserves distance or not.
*Answers will vary. Dilation does not preserve distance.
A dilation is a transformation of size, not position.*



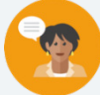
Consider providing a sentence stem for this prompt:

A dilation _____ an isometry because an isometry preserves _____ and a dilation _____ preserve _____.

1.5 Transformations on the Plane – Part 1

Lesson Introduction

 Benchmark	MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.		 Learning Targets	- I can identify the angle of rotation of a transformation of a figure on a plane. - I can identify the center of rotation of a transformation of a figure on a plane. - I can identify the direction of a rotation of a transformation of a figure on a plane. - I can distinguish a rotation from other transformations.	
 Benchmark Coherence	Building On			Working Toward	
	MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.			N/A	
 Essential Questions	- How can the center of rotation be found? - Is rotation a rigid transformation? - Is there more than one way to define the rotation of a preimage to image?				
 Lesson Narrative	Throughout this lesson, students observe and measure rotations using different methods. By creating a rotation, reflection, and translation of the same preimage, comparisons between the differences of each transformation are made through the lens of an Error Analysis. Students engage in determining the direction, center, and degree of rotation of multiple transformations.				
 Component Summary	1.5.1 Warm-Up (~10 min.)	Creating images of different views (SE p. 28)	1.5.3 Guided Practice (5-10 min.)	Describing rotations (SE p. 32)	
	1.5.2 Guided Instruction (25 min.)	Describe transformations on plane (SE p. 29)	1.5.4 Wrap-Up (~5 min.)	Describing rotations (SE p. 33)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - N/A <u>Additional Glossary:</u> - Clockwise - Counterclockwise - Degree		- Patty Paper - Ruler - Protractor (Optional) Markers or Colored Pencils		For 1.5.2 Guided Instruction, consider providing a copy on patty paper of the preimage and point of rotation for students who struggle with tracing precisely.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse clockwise and counterclockwise. - Students may enter this lesson with the idea that a reflection is the same as a 180° rotation. - Students may incorrectly use the protractor to measure the angles.	N/A
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share There are a variety of opportunities for students to Think-Pair-Share throughout 1.5.2 Guided Instruction. Students need the opportunity to think through each of the discussion questions, process the information in pairs, and share their results to the class.	MA.912.MTR.5.1: Patterns and Structure Students relate what they previously learned in grade 8 to further understand rotations. Students have to create plans and logically order steps to determine the center of rotation, the degree of rotation, and the direction of rotation. Students will look for similarities amongst the rotations to help understand why many rotations can be described in more than one way.
 Differentiate Instruction	Students have an opportunity to physically engage with the measurement of angles in 1.5.2 Guided Instruction. This is an opportunity to encourage students to play with the different angles and possibly test each other on angles. Students could draw angles and ask partners to find the measure of the angles.	
Lesson Notes:		

1.5.1 Warm-Up: Rotations in Drawings

Component Overview: Students consider how images will be drawn and seen from different views.



1.5.1 Warm-Up: Rotations in Drawings

The picture shows the front, side, and back view of an animated character.



1. What do you notice about all three views — front, side, and back, of the animated character?

Answers will vary. I notice that it is the same person, just turned around.

When animators and cartoonists sketch the movement of a character, they use a sheet like the one shown.

Front	Side	Back



Students are encouraged to provide any answer they can think of. The way students see this variety of perspectives could help students understand the idea of rotation, which will be the focus of the lesson.

1.5.1 Warm-Up: Rotations in Drawings (continued)

2. Use one of the pictures below, or your own idea, to sketch the three views. Use the dotted lines to help you align key features of the picture.



Answers will vary.

Front	Side	Back



Consider speaking with students about how rotations of a character depicts movement in a three-dimensional space.



Students may struggle with the idea of having to draw. Encourage students to try so that they develop an understanding of rotations. Remind students that this is an opportunity to develop a growth mindset.

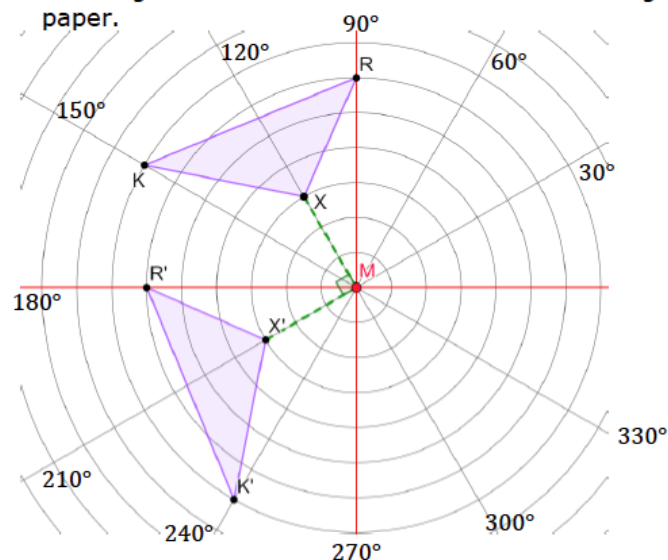
1.5.2 Guided Instruction: Transformations on the Plane

Component Overview: Students determine the angle of rotation between a preimage and an image using circular grid paper and with a protractor. Students draw and compare the image of a rotation to the image of a reflection and a translation of the same preimage.



1.5.2 Guided Instruction: Transformations on the Plane

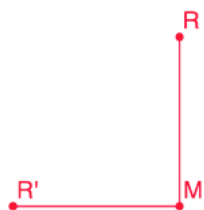
1. The diagram shows $\triangle R K X$ and $\triangle R' K' X'$ on circular grid paper.



Follow the steps below.

- Use a piece of patty paper to copy $\triangle R K X$ and point M .
- Place the patty paper so that it aligns with $\triangle R K X$ and point M .
- Place the point of your writing utensil on point M and rotate $\triangle R K X$ so that it lays on top of $\triangle R' K' X'$.

- a. Draw $\angle R M R'$.



Students often struggle with rotations. By introducing the circular grid paper students can relate the angle measures on the grid to the rotation. It may be helpful to ask students what they notice and wonder about the diagram. Help students notice that the vertices of the preimage and the image are on the same circle. Students can count the number of degrees from preimage vertex to the image vertex or notice that the angle of the vertex X of the preimage, the center of rotation, and vertex X of the image create a right angle.

1.5.2 Guided Instruction: Transformations on the Plane (continued)

- b. Draw $\angle KMK'$.



- c. Complete the table.

Angle	Measurement (degrees)
$\angle XMX'$	90°
$\angle RMR'$	90°
$\angle KMK'$	90°

- d. Discuss with your partner how the angle measures on the circular grid can be used to verify the rotation and how to determine the direction of rotation. Summarize your discussion.

Answers will vary. Subtract the degree measure of the new point minus the degree measure of the original point. The direction depends on if the degree measure increased (counterclockwise) or decreased (clockwise) when moving from the original point to the new point.

- e. Identify the transformation of $\triangle RKX$. Describe the transformation using direction, degrees, or scale factor.

Type of Transformation	Center, Line, or Movement	Description
○ Dilation ○ Rotation ○ Reflection ○ Translation	○ Center of dilation is _____. ○ Center of rotation is <u>M</u>. ○ Line of reflection is _____. ○ Down ○ Left ○ Right ○ Up	<i>The transformation was a 90° rotation counterclockwise around point M.</i>



Students may ask for a protractor here to measure the angles or may provide an estimation. Direct students back to the circular graph to determine the degree measures.



Think-Pair-Share

For question 1d, students could be instructed to begin working independently with about 30 seconds of “think time.” After this, students could work with their partner to discuss the ideas they developed. The pair of students should develop a summary that could be shared with the class.

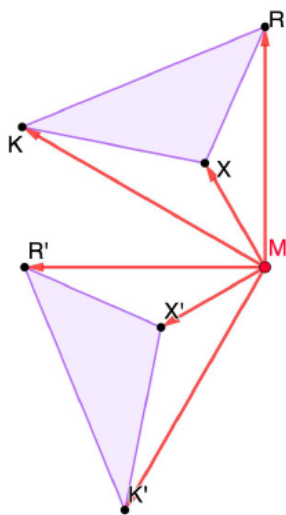
1.5.2 Guided Instruction: Transformations on the Plane (continued)

- f. Discuss with your partner whether or not the transformation could be described as "a rotation of 270° clockwise centered at point M ." Explain why or why not.

Yes, moving in the other direction would be "a rotation of 270° clockwise centered at point M " because all of the vertices of triangle RKX will map onto triangle $R'K'X'$..

2. The diagram shows $\triangle RKX$ and $\triangle R'K'X'$ without the circular grid paper.

- a. Draw $\angle XMX'$, $\angle RMR'$, and $\angle KMK'$.



- b. Discuss with your partner how the degree of rotation can be determined using the angles that connect the vertices and the center of rotation.

Answers will vary.



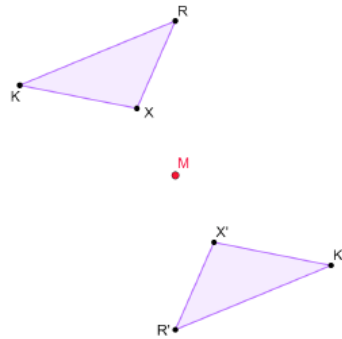
Spend time on question 1f to ensure students understand why rotations can be described in two different ways.



Question 2 is the same set of triangles and rotation as question 1. This helps students build their understanding of rotations without any grid. Help students recognize that the degree, center, and direction of the rotation can be determined without the circular grid.

1.5.2 Guided Instruction: Transformations on the Plane (continued)

3. The diagram shows $\triangle RKKX$ and $\triangle R'K'X'$ using a different transformation.



- a. Identify the transformation of $\triangle RKKX$. Describe the transformation using direction, degrees, or scale factor.

Type of Transformation	Center, Line, or Movement	Description
☐ Dilation ☒ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☒ Center of rotation is <u>M</u>. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	<i>The transformation was a 180° rotation counterclockwise around point M.</i> <i>The transformation was a 180° rotation clockwise around point M.</i>

- b. Two students' descriptions of the transformation of $\triangle RKKX$ are shown.

Michela	Marc
a reflection across line M	a translation down

- Use a piece of patty paper to copy $\triangle RKKX$.
- Perform each student's transformation to prove that each transformation does not work.



Students are given a rotation of 180° . Help students relate that the angle is 180° , because there is a straight line through the center of rotation from each point on the preimage to the image.



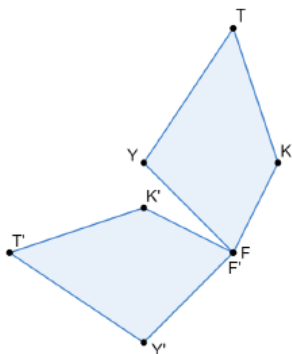
Consider having a student perform the reflection and the translation to demonstrate Michela's and Marc's transformations. Alternately, you can show each using software such as GeoGebra.

1.5.2 Guided Instruction: Transformations on the Plane (continued)

- c. Discuss with your partner why both students are wrong. Write your reasoning for why the transformations do not work.

Answers will vary. Michela is incorrect because her answer would have points R' and K' on opposite sides of X' . Marc is also incorrect because the point X' is below $R'K'$ after his transformation instead of above it.

- d. The diagram shows $TKFY$ and $T'K'F'Y'$.



Identify the transformation of $TKFY$. Describe the transformation using direction, degrees, or scale factor.

Type of Transformation	Center, Line, or Movement	Description
○ Dilation ○ Rotation ○ Reflection ○ Translation	○ Center of dilation is _____. ○ Center of rotation is F. ○ Line of reflection is _____. ○ Down ○ Left ○ Right ○ Up	<i>This is a 90° rotation Counterclockwise about point F.</i>



Students may struggle understanding why the reflection is not correct. Specifically point out that the vertices do not align.



For a visual entry-point, consider having students use colored markers to draw the angles from preimage point to image point. Then, using a protractor, students can measure the degree of rotation.



When giving descriptions of the answers always include the alternate rotation.

1.5.3 Guided Practice: Rotations

Component Overview: Students determine the direction of rotation, center of rotation, and degree of rotation for three different transformations.



1.5.3 Guided Practice: Rotations

The table shows three different rotations of $TKFY$. Determine the direction of rotation, center of rotation, and the degree of rotation for each.

Transformation	Direction, Center, and Degree
	- Direction of rotation is <u>Clockwise</u>. - Center of rotation is <u>P</u>. - Degree of rotation is <u>90°</u>.
	- Direction of rotation is <u>Clockwise</u>. - Center of rotation is <u>T</u>. - Degree of rotation is <u>90°</u>.
	- Direction of rotation is <u>Clockwise</u>. - Center of rotation is <u>P</u>. - Degree of rotation is <u>180°</u>.



To add complexity, students can determine the degree of rotation for each transformation if rotated both clockwise and counterclockwise.

1.5.4 Wrap-Up: Ticket Out the Door

Component Overview: Students complete a Ticket Out the Door by identifying the direction, center, and degree of each rotation.



1.5.4 Wrap-Up: Ticket Out the Door

1. The table shows two different rotations of $TKFY$. Determine the direction of rotation, center of rotation, and the degree of rotation for each.

Transformation	Direction, Center, and Degree
	- Direction of rotation is Counterclockwise. - Center of rotation is K. - Degree of rotation is 180°.
	- Direction of rotation is Counterclockwise. - Center of rotation is P. - Degree of rotation is 90°.

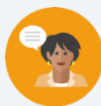


Use student understanding to inform your instruction and to determine where in future lessons and units you can help students build a better understanding of rotations. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.

1.6 Transformations on the Plane – Part 2

Lesson Introduction

	Benchmark	MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.			Learning Target	I can describe a single transformation that was applied.
	Benchmark Coherence	Building On MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.		Working Toward N/A		
	Essential Questions	- What are differences between the transformations? - How can each transformation be identified? - What is the simplest way to describe each transformation?				
	Lesson Narrative	This lesson is a culmination of all the transformations in the previous lessons. Students progress through the lesson by identifying and describing the transformation that results in an image from a given preimage. Students are led to being able to identify and describe all transformations.				
	Component Summary	1.6.1 Warm-Up (~5 min.)	Which polygon does not belong? (SE p. 34)	1.6.3 Guided Practice (5 min.)	Describing a transformation (SE p. 37)	
		1.6.2 Guided Instruction (25 min.)	Identifying and describing single transformations (SE p. 35)	1.6.4 Your Turn! (5 min.)	Describing a transformation (SE p. 39)	
		1.6.5 Wrap-Up (~10 min.)	Identifying and describing single transformations (SE p. 40)			
	Resources	Glossary		Materials		Additional Preparation
		<u>Key Terms:</u> - N/A <u>Additional Terms:</u> - Translation - Rotation - Reflection - Dilation		(Optional) Patty Paper (Optional) Ruler		N/A

 Teacher Tips	Common Misconceptions Students may believe that all transformations are rigid transformations.	Things to Consider - If students are struggling with finding the point of dilation on 1.6.2 Guided Instruction question 4, consider starting with images where the point of dilation is the center of the image. Once this is something students can understand, consider moving it to a point on the edge of the image. Finally, try one where the point is outside of the image. - If time is an issue, then consider assigning 1.6.4 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Number Talk For 1.6.2 Guided Instruction, display each transformation one at a time. Give the students a few minutes to quietly think and give a signal when they have an answer. Select students to share their strategies for how they approached the problem. Press students to provide details they see that have not already been shared.	MA.K12.MTR.1.1: Participate in Effortful Learning Identifying transformations can be a difficult and challenging task. Students will build perseverance by listening to others and modifying their methods for determining a transformation. Having students share their methods will encourage engagement and allow others to make sense of the task.
 Differentiate Instruction	The transformations that are present throughout this lesson starting at 1.6.2 Guided Instruction could be represented with physical objects. This could be patty paper, Tangrams, Play-Doh, or other manipulatives.	
	In 1.6.4 Your Turn, encourage students to discuss alternative methods of getting to the same answer. This can lead pairs or groups to discuss the topic and come up with alternatives that otherwise may not have been thought of.	
Lesson Notes:		

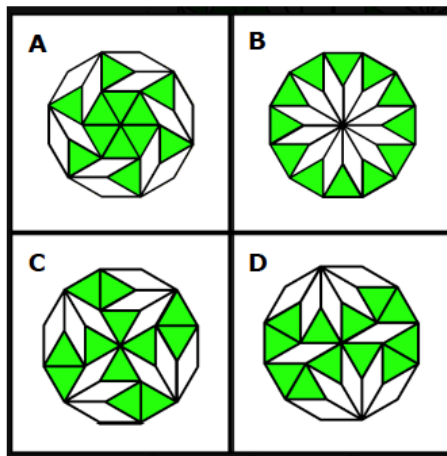
1.6.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students identify which polygon they believe does not belong. Each polygon has a reason why it does not belong. The selection students make is not important, but the reason why is what should be highlighted.



1.6.1 Warm-Up: Which One Doesn't Belong?

1. The diagram shows four polygons.



- a. Which polygon does not belong?

Answers will vary.

- b. Describe your reasoning.

Answers will vary. Polygon A has a perfect hexagon. Polygon B has all white spaces sharing a line. Polygon C has no more than three shapes sharing a side. Polygon D looks as if it has a mirror in the middle.



There is no right or wrong answer. Help students understand that the purpose is for them to consider which does not belong to defend their choice. Encourage students to have flexible thinking.



Students could be challenged to stretching their reasoning to explain why another option could also be the answer. This would allow students to consider another perspective that they may have not seen before.

1.6.2 Guided Instruction: Determining a Single Transformation

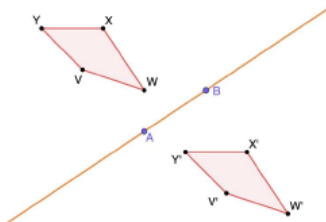
Component Overview: Given a preimage and an image, students identify and describe the type of transformation used. Students will complete a chart of how they distinguish between the four transformations.



1.6.2 Guided Instruction: Determining a Single Transformation

1. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.

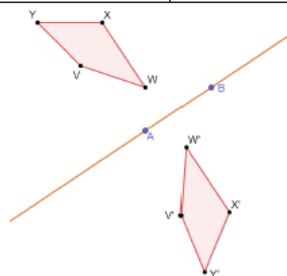
Describe the single transformation that was used to create $V'W'X'Y'$.



Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☒ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☒ Down ☐ Left ☒ Right ☐ Up	

2. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.

Describe the single transformation that was used to create $V'W'X'Y'$.



Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☒ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☒ Line of reflection is $\overline{AB}$. ☐ Down ☐ Left ☐ Right ☐ Up	



Students may benefit from having a piece of patty paper to help them determine the transformation. For many students, visualization of transformations is difficult and should be supported with tools, such as patty paper.



If students choose the incorrect transformation, consider asking the following question.

- What do you notice about the distance between each point of the preimage to its corresponding point in the image?
- If this were a reflection or rotation, where would vertex W be in relation to the line segment or the points?

If students are correct, asking these questions will help them defend their reasoning.



If students choose the incorrect transformation, consider asking the following question.

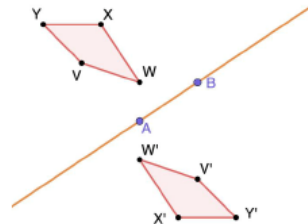
- What do you notice about the orientation of the vertex points in the preimage compared to the order of the vertex points in the image?

If students are correct, asking this question will help them defend their reasoning.

1.6.2 Guided Instruction: Determining a Single Transformation (continued)

3. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.

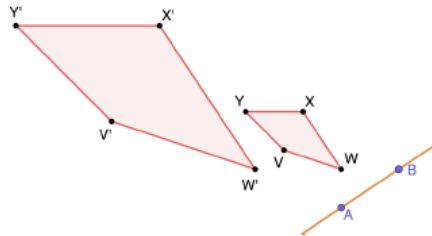
Describe the single transformation that was used to create $V'W'X'Y'$.



Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☒ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☒ Center of rotation is A . ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	180° rotation clockwise or counterclockwise

4. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.

Describe the single transformation that was used to create $V'W'X'Y'$.



Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☒ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☒ Center of dilation is B . ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	Scale factor of 2.5



If students are having a hard time finding the point of rotation, students can draw line segments connecting each point of the preimage to its corresponding point on the image. Where the four lines intersect is the point of rotation.

To verify a point of rotation, students can place the point of a compass on a possible "point of rotation." If the point is correct, the compass can be opened to any specific point of the preimage and an arc drawn that will perfectly intersect with the corresponding point on the image.



Students may need to be reminded that rotation can be clockwise or counterclockwise.



To find the point of dilation, students should draw lines extending through each point of the preimage and the corresponding point on the image. The point of dilation is where these four lines intersect.

Students can determine the scale factor by finding the actual measurements or estimating using patty paper.

1.6.2 Guided Instruction: Determining a Single Transformation (continued)

5. Complete the chart so it can be used to distinguish between similar-looking transformations. First, write your own hints. Then, exchange hints with your partner.

Transformation	My Hints	My Partner's Hints
Dilation	<i>Answers will vary.</i>	<i>Answers will vary.</i>
Reflection	<i>Answers will vary.</i>	<i>Answers will vary.</i>
Rotation	<i>Answers will vary.</i>	<i>Answers will vary.</i>
Translation	<i>Answers will vary.</i>	<i>Answers will vary.</i>



When monitoring student responses to question 5, make note of students who have different hints. Call on these students to share their hints with the whole class. Students benefit from hearing the thinking of others.

6. Which transformations resulted in congruent figures?

- ☐ Dilation
- ☒ Reflection
- ☒ Rotation
- ☒ Translation

7. When a transformation results in a congruent figure, what is another name that can be used for that transformation?

The transformation that results in a congruent figure is a rigid transformation.



Another name for a transformation that results in a congruent figure is an isometry.

1.6.3 Guided Practice: Describing Transformations

Component Overview: Students identify and describe each transformation that results in an image from a given preimage.

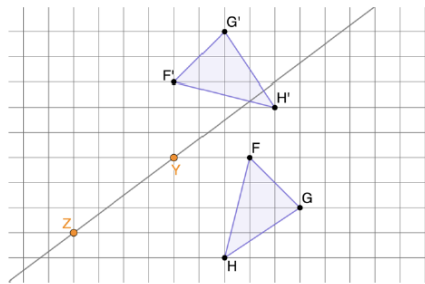


1.6.3 Guided Practice: Describing Transformations

Each of the diagrams shows $\triangle FGH$ and $\triangle F'G'H'$. $\triangle F'G'H'$ is a transformation of $\triangle FGH$.

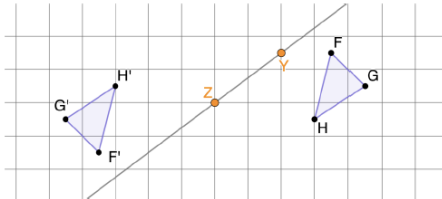
- Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.

Rotate 90° counterclockwise around point Y or Rotate 270° clockwise around point Y



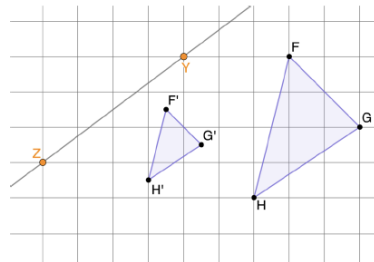
- Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.

Rotate 180° counterclockwise around point Z.



- Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.

Dilation with scale factor of $\frac{1}{2}$ and center of dilation point Z.



- Review your hints chart and add anything else you think will be helpful.

Answers will vary.



Allow informal observation throughout the 1.6.2 Guided Instruction to dictate how “guided” 1.6.3 Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.



Consider challenging students to find another way to end up with the same post-image without using rotation.



How can finding the slope between the center of rotation and a point on the preimage be compared to the slope between the center of rotation and the corresponding point on the image to find the scale factor?

1.6.4 Your Turn!

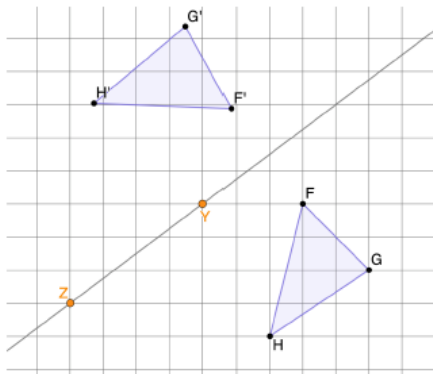
Component Overview: Students identify and describe each transformation that results in an image from a given preimage.



1.6.4 Your Turn!

1. The diagram shows $\triangle FGH$ and $\triangle F'G'H'$. $\triangle F'G'H'$ is a transformation of $\triangle FGH$.

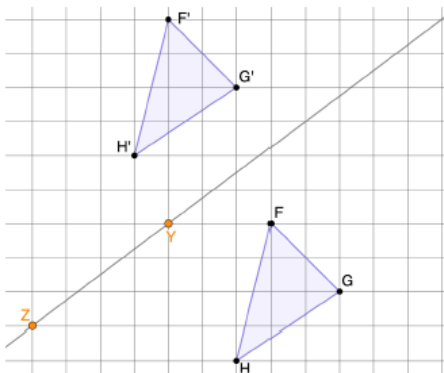
Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.



The transformation that was applied was a reflection over line $\overleftrightarrow{YZ}$.

2. The diagram shows $\triangle FGH$ and $\triangle F'G'H'$. $\triangle F'G'H'$ is a transformation of $\triangle FGH$.

Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.



The transformation was a translation of six units up and three units left.



Monitor while students are working to determine if there are any questions that need to be debriefed before moving to this component. This could include highlighting student voices to share rich discussions or comments overhead while you circulated or to address common misconceptions that were evident among the majority of the class.

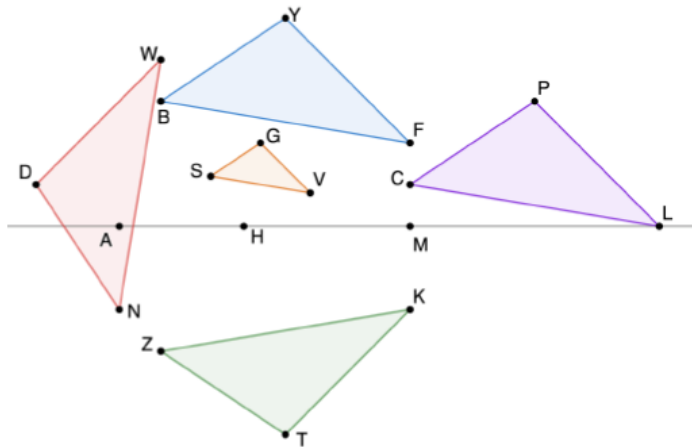
1.6.5 Wrap-Up: Ticket Out the Door

Component Overview: Students summarize their learning by identifying and describing the transformation that will produce each image.



1.6.5 Wrap-Up: Ticket Out the Door

- The diagram shows four different single transformations of $\triangle YFB$.



Complete the chart by naming and describing the type of transformation that was applied to $\triangle YFB$ to result in each triangle.

Triangle	Type of Transformation	Description of the Transformation
$\triangle PLC$	Translation	The transformation was a slide down and shifted to the right.
$\triangle TKZ$	Reflection	It was a reflection over line $\overline{HM}$.
$\triangle GVS$	Dilation	The triangle has a center of dilation at point H.
$\triangle DWN$	Rotation	The rotation is 90° counterclockwise at point H.



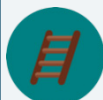
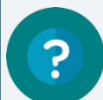
The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.



Be careful to check each type of transformation along with the description. Students may see other paths to the correct answer.

1.7 Congruence and Similarity

Lesson Introduction

 Benchmark	MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.		 Learning Targets	- I can recognize the relationship between corresponding sides and corresponding angles of congruent and similar figures. - I can write congruence statements to relate corresponding parts following a rigid transformation. - I can write similarity statements to relate corresponding parts following a dilation.
	MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar.			
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What determines if a shape is congruent? - What determines if a shape is similar? - How do I write a congruence statement? - How do I write a similarity statement?			
 Lesson Narrative	Students engage in identifying and describing transformations to determine which ones produce congruent figures and which ones produce similar figures. Definitions of congruent and similarity are presented, and students determine congruent and similar parts of transformed figures.			
 Component Summary	1.7.1 Warm-Up (~5 min.)	The time needed to become an expert (<i>SE p. 41</i>)	1.7.4 Bring It Together (5 min.)	Defining congruent and similarity (<i>SE p. 45</i>)
	1.7.2 Collaboration (10 min.)	Transforming a line segment to produce a congruent or similar figure (<i>SE p. 41</i>)	1.7.5 Guided Instruction (10 min.)	Identifying transformations that produce congruent and similar images (<i>SE p. 46</i>)
	1.7.3 Guided Practice (10 min.)	Transforming angles to produce congruent angles (<i>SE p. 44</i>)	1.7.6 Wrap-Up (~5 min.)	Reflection on what was learned, what was new, and what is unclear (<i>SE p. 47</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Congruent - Similarity <u>Key Terms:</u> - N/A		- Patty Paper - Ruler (Optional) Protractor	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students do not recognize that all line segments are transformations of a preimage.	Consider if students will need to identify all congruent figures as similarities of 1:1 in your classroom. Set up the expectation early to avoid confusion and misconceptions.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns Throughout the 1.7.2 Collaboration, encourage students to engage in the Take Turns routine. The partner who responds with the answer first should alternate for each question. Students must explain the rationale for their answer choice and listen to the rationale of their partner. Taking turns gives students opportunities to construct arguments and to critique the reasoning of others.	MA.K12.MTR.4.1: Discussion and Reflection In 1.7.2 Collaboration, students will work together to determine and describe transformations that produce an image from a preimage. After working together, the students share their thinking with other classmates. Students also reflect on the arguments of two sample students to determine transformations that produce congruence or similarity.
 Differentiate Instruction	Students should have an opportunity to engage with the Key Terms, similar and congruent. This will be accomplished both visually and kinesthetically if you challenge students to define each word using only pictures. Consider having students add these terms to a Word Wall. Ask students to read through these definitions and create an artistic representation of each word. The artistic approach will help students explore and retain definitions and key differences between congruence and similarity.	
Lesson Notes:		

1.7.1 Warm-Up: 10,000 Hours to Awesome

Component Overview: Students read research about the amount of time it takes for someone to become an expert at something.



1.7.1 Warm-Up: 10,000 Hours to Awesome

Best-selling author Malcolm Gladwell states in his book "Outliers" that it takes 10,000 hours of practice to be a phenom — so freakishly awesome or great at something. In 1973, Herbert Simon and William Chase wrote a paper published in *American Scientist*. Simon and Chase did experiments with chess-playing tasks and computer simulations to determine how long it would take someone to reach the grandmaster level in the game of chess. Their conclusion was that it could take as many as 50,000 hours staring at chess positions to become a chess grandmaster.

1. What skill, hobby, or activity would you like to become freakishly awesome at?

Answers will vary. I would like to become freakishly awesome at Ice Hockey.

2. How many years is 10,000 hours?

10,000 hours is just over one year, 1.14 years.

3. How many years is 50,000 hours?

50,000 hours is 5.71 years.



Consider taking the time to log student responses to question 1 so that you get to know them better.

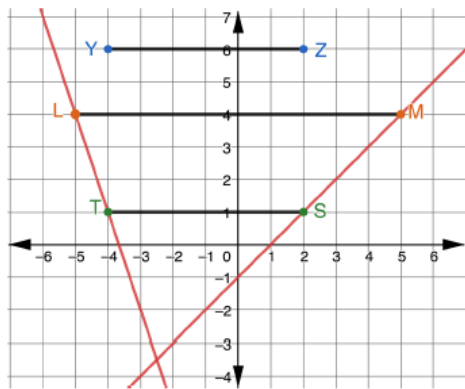
1.7.2 Collaboration: Line Segments

Component Overview: Students begin by identifying the relationship that the length of three line segments have to each other. Students then determine which transformation may have produced the different lines shown and analyze statements from others about the transformations.



1.7.2 Collaboration: Line Segments

1. $\overline{LM}$, $\overline{TS}$, and $\overline{YZ}$ are shown on the coordinate grid.



- What do you notice about the three line segments?
Answers will vary. They are all parallel. Two of the line segments are the same length.
- List at least two different ways to show that the length of $\overline{TS}$ is equal to the length of $\overline{YZ}$.
I could count the squares or measure with a ruler.
- What is the ratio of the length of $\overline{TS}$ to the length of $\overline{LM}$?
 $6:10 = 3:5$
- What is the ratio of the length of $\overline{TS}$ to the length of $\overline{YZ}$?
 $1:1$
- Describe a single transformation that could have been used to transform $\overline{TS}$ to create $\overline{LM}$.
A single transformation to transform $\overline{TS}$ to create $\overline{LM}$ would be a dilation with a scale factor of $\frac{5}{3}$.
- Draw a line that extends through the coordinate plane and passes through points L and T .
Shown on coordinate grid.



Students may make errors during 1.7.2 Collaboration, including writing the scale factor incorrectly. Allow the opportunity for them to correct their own thinking as they work with their partner and compare their work with another partner pair in question 1j. Part of what students may realize by the end of 1.7.2 Collaboration is that the endpoints of a line segment do not have to be in the same order to consider congruence or similarity. Do not be afraid to let the mistakes happen. Wait to discuss them until the transition to the next component. Monitor while students are working to determine if there are any questions that need to be debriefed before moving to 1.7.3 Guided Practice. This could include highlighting student voices to share rich discussions or comments overhead while you circulated or to address common misconceptions that were evident among the majority of the class.



Keep in mind that some students learn ratio notation as $a:b$, a to b , or $\frac{a}{b}$.



While circulating listen for students who may incorrectly state the scale factor as $\frac{3}{5}$ instead of $\frac{5}{3}$. Address this misconception so that it is not carried forward in their learning. Consider the following questions:

- Which line segment is the preimage and which is the image in question 1e?
- Is the preimage smaller or larger than the image?

1.7.2 Collaboration: Line Segments (continued)

- g. Draw a line that extends through the coordinate plane and passes through points M and S .
Shown on coordinate grid.
- h. Where do the two lines intersect?
The intersection is at the point $(-2.5, -3.5)$.
- i. What is a single transformation that could have been used to transform $\overline{TS}$ to create $\overline{YZ}$?
Answers will vary. The transformation could be a translation up by 5. The transformation could be a reflection over the line $y = 3.5$. The transformation could be a 180° rotation at the point $(-1, 3.5)$.
- j. Ask two classmates the transformation they selected. Record their transformation and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
Answers will vary.

Transformation	My Summary of Their Reason	Initials

2. Day’Ron and Zeb were discussing how to transform $\overline{TS}$ to create $\overline{YZ}$.

Day’Ron	Zeb
The only two single transformations are a translation of up 5, or a reflection across the line $y = 3.5$.	There are three single transformations. A translation of up 5, a reflection across the line $y = 3.5$, and a rotation of 180° .



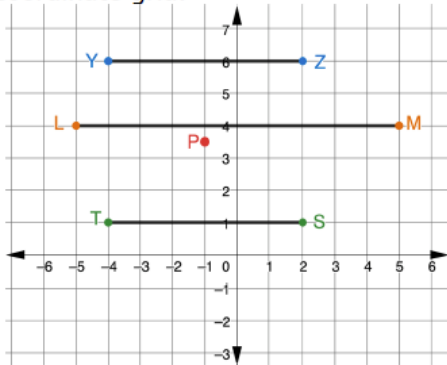
To provide organization to question 1j, consider having students “pair up” by finding “Soulmates.” Students will pair up with another student that has on shoes that match theirs. What defines a match is the student’s own choice. It could be size, color, style, or anything the pair agrees on. After recording their “Soulmate’s” reasoning, students repeat this for finding a second “Soulmate.”



Students may benefit from having patty paper to explore the transformations that are discussed by Day’Ron and Zeb.

1.7.2 Collaboration: Line Segments (continued)

Day'Ron disagreed with Zeb. To prove his point, he used a graphing utility to show $\overline{TS}$, $\overline{YZ}$, $\overline{LM}$ and point P on the coordinate grid.



Day'Ron said, "It is not possible to rotate $\overline{TS}$ 180° so that point T aligns with point Y and point S aligns with point Z ."

Zeb replied, "Yes, but that does not matter because the letters do not have to match exactly. If you rotate $\overline{TS}$ 180° about point P it will coincide with $\overline{YZ}$."

Discuss Day'Ron and Zeb's justifications with your partner. Summarize your discussion.

Answers will vary. The resulting line segment if you rotate $\overline{TS}$ 180° about point P will coincide with $\overline{YZ}$, so it is possible for a 180° rotation about point P to be the transformation used.

3. Which transformation of $\overline{TS}$, $\overline{YZ}$, and $\overline{LM}$ showed congruence?

Transformations to or from $\overline{TS}$ to $\overline{YZ}$ or $\overline{YZ}$ to $\overline{TS}$ are congruent.

4. Which transformation of $\overline{TS}$, $\overline{YZ}$, and $\overline{LM}$ showed similarity?

Transformations involving $\overline{LM}$ are going to be similar.



To frontload for the next activity, ask students to begin discussing the differences between congruence and similarity.

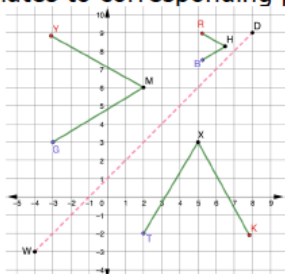
1.7.3 Guided Practice: Angles

Component Overview: Students are asked to describe a variety of transformations. Then, creating images of an angle, students determine that all transformations preserve angle measure.

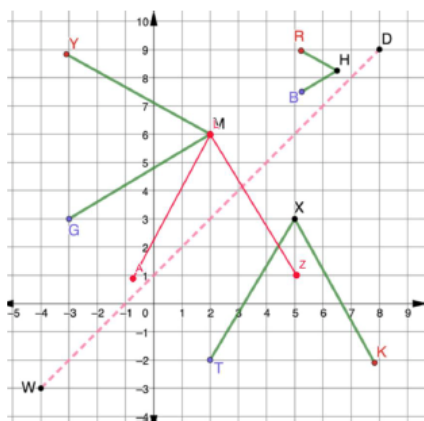


1.7.3 Guided Practice: Angles

$\overline{WD}$, $\angle YMG$, $\angle RHB$, and $\angle KXT$ are shown on the coordinate grid. Each color relates to corresponding points.



- Describe a single transformation that could have been used to transform $\angle YMG$ to create $\angle KXT$.
A reflection over $\overline{WD}$.
- Describe a single transformation that could have been used to transform $\angle YMG$ to create $\angle RHB$.
A dilation centered at point D.
- Use patty paper to create $\angle ALZ$ by rotating $\angle YMG$ 90° counterclockwise centered on point M.



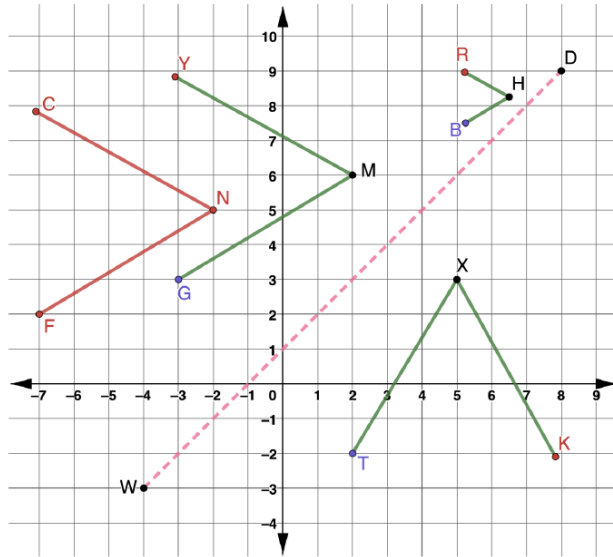
1.7.3 Guided Practice could be considered a guided discovery. Students are led to an understanding that all transformations preserve angle measure. Guiding this whole-group will help you monitor misconceptions and to ensure students understand that an angle's measure is not dependent on the length of the line segments.



Students move from transformations of line segments to transformations of angles. Allow students time to notice that the preimage and image of the angles are congruent. Determine if it is best for your students to pause after question 2 to ask students if the angles on the grid are congruent. Ensure that students do not "eyeball" it and instead use patty paper or a protractor to confirm congruence.

1.7.3 Guided Practice: Angles (continued)

4. Use patty paper to create $\angle CNF$ by translating $\angle YMG$ left 4 down 1.



5. Which angles have the same measure as $\angle YMG$?
Angles $\angle KXT$, $\angle RHB$, $\angle ALZ$, and $\angle CNF$ all have the same angle measure as $\angle YMG$.
6. Which transformation(s) resulted in an angle whose measurement is the same as $\angle YMG$?
All transformations result in the same angle measure: rotation, reflection, translation, and dilation.



Students should use the image copied on the patty paper to confirm that $\angle RHB$ and $\angle TXK$ both have the same angle measure as $\angle YMG$.

1.7.4 Bring It Together: Congruence and Similarity

Component Overview: Definitions of congruent and similarity.



1.7.4 Bring It Together: Congruence and Similarity

1. Complete the statements.

a. A rigid transformation results in a line segment that

- ☒ is
☐ is not

congruent to the original line segment.

b. A dilation results in a line segment that
congruent to the original line segment.

- ☐ is
☒ is not

c. A rigid transformation results in an angle that

- ☒ is
☐ is not

congruent to the original angle.

d. A dilation results in an angle that
congruent to the original angle.

- ☒ is
☐ is not



Two figures are **congruent** if they have exactly the same shape and size. Equivalently, two figures are congruent if one figure can be mapped to the other using a single rigid motion, or a sequence of transformations that includes only rigid motions.



Similarity is having exactly the same shape but not necessarily the same size. Two figures are similar if and only if one figure can be obtained from the other by a dilation, or a sequence of transformations that includes a dilation.



Students will further explore a dilation of 1 and that it results in congruence in question 2. If students chose “is” in the second box, then consider asking them to explain.



To lead students through the process of generating a pictorial definition, consider asking students to underline the keywords of the definitions. This should reduce the number of words needed to define each word. After this, generate a picture of as many keywords as they can. Once they have these pictures, students should attempt to summarize these pictures, or use them to “create a scene” that would define the Key Term.

1.7.4 Bring It Together: Congruence and Similarity (continued)

Two congruent figures can also be considered similar with a scale factor of 1. Although this is known, it is not often required to state that two congruent figures are also similar.

2. Which statement in Question 1 would be false for a dilation with a scale of factor of 1?
Statement b would be false for a dilation with a scale factor of 1 would result in a congruent line segment.

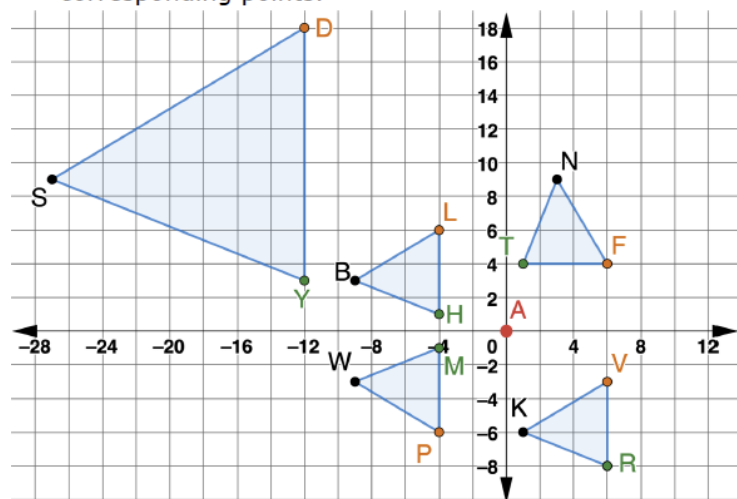
1.7.5 Guided Instruction: Congruence and Similarity Statements

Component Overview: Students look at the variety of figures and identify the transformation that could create each image, and if the images are congruent or similar to the preimage. Students then identify congruent or similar parts of the preimage and image.



1.7.5 Guided Instruction: Congruence and Similarity Statements

1. Five triangles are shown on the coordinate grid. $\triangle FNT$, $\triangle VKR$, $\triangle PWM$, and $\triangle DSY$ are created using a single transformation of $\triangle LBH$. Each color relates to corresponding points.



- a. Complete the table.

Triangle	Description of Transformation	Congruent or Similar
$\triangle FNT$	Rotation 90° Clockwise about point A	Congruent
$\triangle VKR$	Translation down 9 and to the right 10.	Congruent
$\triangle PWM$	Reflection across the x -axis.	Congruent
$\triangle DSY$	Dilation centered at point A with a scale factor of 3.	Similar



Take the time to show that the corresponding sides of the triangles are congruent or similar and that the corresponding angles are congruent. This lesson built up from line segments and angles to a triangle, so breaking the triangles to their parts lays the foundation for students to understand that the congruence or similarity of an image to its preimage can be considered by relating the corresponding line segments and angles.



Students may benefit from having patty paper or another manipulative that can be used to move $\triangle LBH$ to discover and describe the transformation.

1.7.5 Guided Instruction: Congruence and Similarity Statements (continued)

- b. Use the information about the transformations to complete each statement. *Answers may vary.*

$$\overline{LH} \cong \overline{MP}$$

$$\overline{TN} \cong \overline{HB}$$

$$\overline{YS} \sim \overline{BH}$$

$$\angle DYS \cong \angle LHB$$

$$\overline{MP} \sim \overline{YD}$$

$$\overline{FT} \cong \overline{LH}$$

$$\overline{PW} \cong \overline{BL}$$

$$\angle PMW \cong \angle LHB$$

$$\angle VKR \cong \angle LBH$$

- c. Discuss with your partner why there could be multiple answers to each statement. Summarize your discussion.
Answers will vary.



For a challenge, ask students to list all possible answers for each object in question 1b.

1.7.6 Wrap-Up: Cool Down

Component Overview: Students reflect on what they learned, what they did not know, and a question they still have.



1.7.6 Wrap-Up: Cool Down

Complete the statements about today's lesson.

1. Something I learned today ...

Answers will vary.

2. I didn't know ...

Answers will vary.



Answers to complete the statement "One question I still have ..." is where students may be asking for help, even if they do not say anything during class.

3. One question I still have ...

Answers will vary.

1.H1 Symmetries

Lesson Introduction

 Benchmark	MA.912.GR.2.4 Determine symmetries of reflection, symmetries of rotation, and symmetries of translation of a geometric figure.		 Learning Targets	- I can determine symmetries of reflection. - I can determine symmetries of rotation. - I can determine symmetries of translation.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- What is symmetry? - What symmetries can be used to create a tessellation?			
 Lesson Narrative	Students observe and discuss how transformations can be used to show symmetry. Following a pattern, students create and identify translational, reflectional, and rotational symmetries. Connections are made between the number of lines of symmetry with the order of rotational symmetry a polygon has. Tessellations are created using a pattern chosen by the individual students.			
 Component Summary	1.H1.1 Warm-Up (~5 min.)	Notice and Wonder about M.C. Escher artwork (<i>SE p. 48</i>)	1.H1.3 Activity (20 min.)	Creating a tessellation (<i>SE p. 51</i>)
	1.H1.2 Exploration (20 min.)	Describing and creating translational, reflectional, and rotational symmetries (<i>SE p. 48</i>)	1.H1.4 Wrap-Up (~5 min.)	Classifying transformations (<i>SE p. 52</i>)
 Resources	Glossary		Additional Preparation	
	<u>Key Terms:</u> - Tessellation <u>Additional Terms:</u> - Symmetry		After presenting the definition of tessellation in 1.H1.2 Exploration, consider projecting images of other tessellations, especially famous ones by M.C. Escher.	
			Materials 3 x 5 Note Cards 8.5 x 11 Paper - Scissors (Optional) Colored Pencils (Optional) Tangrams or Polygon Shapes	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may only see one possible type of transformation. - Students may confuse horizontal and vertical cuts.	- This lesson should be taught after Lesson 7 for Honors courses. - Consider instilling “stops” in the lesson to check in with partner pairs and their understanding. Suggested stops are included. - Rotational symmetry is a new concept. Consider doing question 7 in 1.H1.2 Exploration as a whole group. - Consider allowing students to complete 1.H1.3 Activity for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In 1.H1.1 Warm-Up, students write things they wonder about two images of M.C. Escher’s artwork. This activity sets the stage for students to look at transformational symmetries and tessellations.	MA.K12.MTR.5.1: Patterns and Structure Students will develop a tessellation in 1.H1.3 Activity. This image will emphasize the importance of following a pattern. Students will be spending time focusing on relevant details in their pattern and creating plans to logically produce an overall image. This activity helps students recognize patterns in the world around them and make connections to mathematical concepts.
 Differentiate Instruction	Students will use repeated translations to develop a tessellation pattern in 1.H1.3 Activity. This pattern will continue across an entire page and should end up as a clearly artistic representation of the mathematical concepts presented throughout Unit 1. The process of cutting and taping the 3 x 5 note cards will allow students to have a great kinesthetic connection with the topic. Once the tessellation is being developed, the aesthetics of the pattern will begin to speak to students' visual learning needs. This emphasis can be amplified by the addition of color and implementation of other key elements from other courses.	
Lesson Notes:		

1.H1.1 Warm-Up: Artwork of M.C. Escher

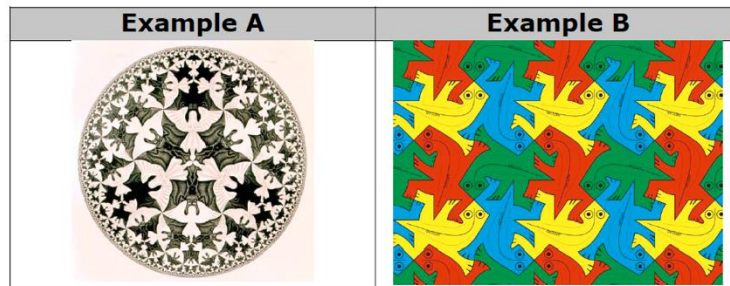
Component Overview: Students view two images of M.C. Escher's artwork, writing anything they notice or wonder. Then, students state the similarities and differences in the images.



1.H1.1 Warm-Up: Artwork of M.C. Escher

M.C. Escher (1898-1972) was a Dutch artist whose work was inspired by mathematics. Though initially trained as an architect, he became one of the most famous and influential graphic artists in the world.

Below are two examples of Escher's work.



Consider having students write down two things they notice about each of the examples of Escher's work. Then, give students a chance to turn and talk about what they initially noticed about the artwork. Students can then complete question 1.

1. Describe any similarities and differences you notice about the two examples.

Similarities	Differences
Answers will vary. Both pieces have images that are repeated. In both pieces, the repeated images are rotated.	Answers will vary. Example A is black and white, and Example B is in color. In Example B, all the geckos are the same size, but in Example A, the bats and angels are different sizes.

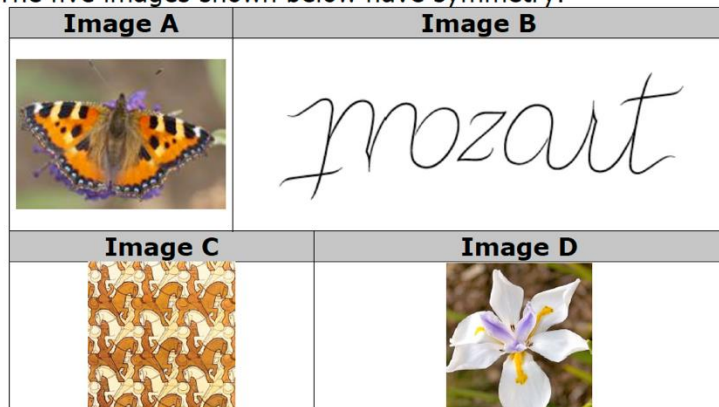
1.H1.2 Exploration: Symmetry

Component Overview: Students determine which of five images have symmetry and what transformations show that one part of the image aligns with the other part(s) of the image. Students identify tessellations that were created using translational, reflectional, and rotational symmetry. A table is completed with the number of lines of symmetry and the order of rotational symmetry for several polygons.



1.H1.2 Exploration: Symmetry

The five images shown below have symmetry.



Students may struggle most with Image B and Image C. Avoid providing any hints prior to working through the remainder of this component. Revisit these, and any other images, before moving on to 1.H1.3 Activity.

- Discuss with your partner why each image has symmetry and which transformations could be used to show that one part of the image aligns with the other.
Answers will vary. They are either reflections or rotations.
- Complete the table. Some images may align with more than one type of transformation.

Image	Transformations
A	☐ reflection across a line ☐ rotation about a point ☐ translation
B	☐ reflection across a line ☐ rotation about a point ☐ translation
C	☐ reflection across a line ☐ rotation about a point ☐ translation
D	☐ reflection across a line ☐ rotation about a point ☐ translation

1.H1.2 Exploration: Symmetry (continued)

A **tessellation** is an arrangement of shapes closely fitted together in a repeated pattern without gaps or overlaps.

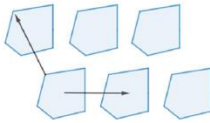
3. Which image is a tessellation?

Image C is a tessellation.

Translational symmetry is when sliding a pattern is used to create symmetry.

Reflectional symmetry is when a pattern can be reflected to create symmetry.

Rotational symmetry is when a rotation can be used to create symmetry. The table shows examples of each type of symmetry.

Image	Symmetry
	Translational symmetry
	Reflectional symmetry
	Rotational symmetry



Consider implementing a “stop” after question 3 and check-in with student learning as a whole group.

1.H1.2 Exploration: Symmetry (continued)

- Micah commented to his partner that he thought the picture for reflectional symmetry could also be translational symmetry. Discuss this with your partner.
- For each pattern, use the first two squares to complete the pattern.

Pattern A

Pattern B

Pattern C

- Work with your partner to determine the type of symmetry each pattern displays.

Pattern	Type of Symmetry
A	<i>Translational symmetry</i>
B	<i>Rotational symmetry</i>
C	<i>Reflectional symmetry</i>



As you circulate consider engaging students in a discussion of why the image represents the symmetry listed. Ask students to include what should be looked for in determining the type of symmetry.



This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

1.H1.2 Exploration: Symmetry (continued)

The order of rotational symmetry of a shape is the number of times it can be rotated around a full circle and still look the same.

7. Complete the chart for each polygon.

Polygon	Figure	Number of Lines of Symmetry	Order of Rotational Symmetry
Isosceles triangle		1	1
Equilateral triangle		3	3
Square		4	4
Kite		1	1
Regular Pentagon		5	5
Regular Hexagon		6	6



For a kinesthetic entry-point, provide a copy of the image that can be manipulated to determine the type of transformation prior to the students trying to draw the third and fourth images.

1.H1.3 Activity: Creating a Tessellation

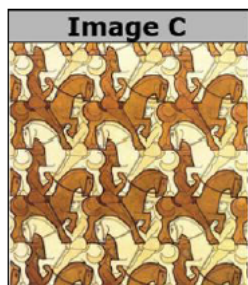
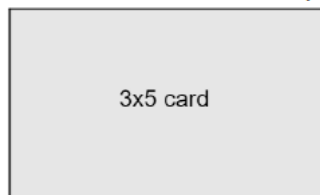
Component Overview: Students create a tessellation.



1.H1.3 Activity: Creating a Tessellation

Image C is a tessellation that was created by MC Escher, a famous Dutch artist.

To create a tessellation, you will need a 3x5 index card and tape.

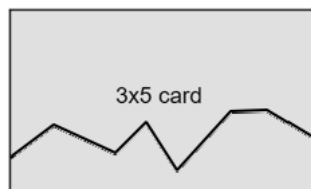


Colored pencils will allow students to add color. This activity can be started in class and completed at home. The teacher can display the completed art.

1. Create your own tessellation by choosing one of the following options. The more cuts you make, the more interesting your pattern will become.

Horizontal Cut

Draw a curve or pattern along a horizontal side of the card.



Vertical Cut

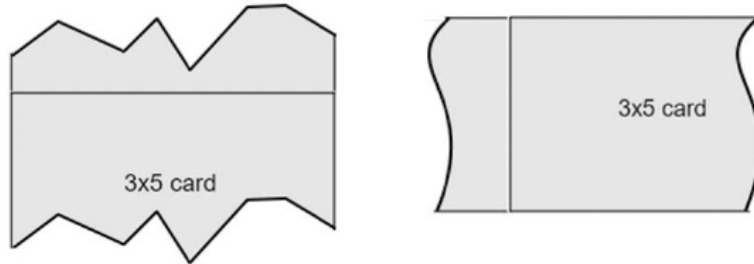
Draw a curve or pattern along a vertical side of the card.



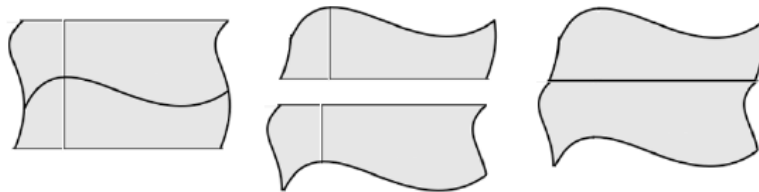
Consider collaborating with an art or graphic design instructor to see if tessellations are in their course guides. Engagement can be increased by giving students real-world meaning through cross-curricular instruction. Students can transfer content to multiple academic disciplines throughout this tessellation activity.

1.H1.3 Activity: Creating a Tessellation (continued)

Cut along the curve or pattern and shift the piece to the opposite side. Tape the piece on.



It is also possible to make multiple cuts as shown.



2. Use your card to create a tessellation by filling up a sheet of paper. Then, color your tessellation.



Students moving through the process of cutting and rejoining the note card can significantly engage students in a kinesthetic way. The addition of color can significantly improve the students' visual connection with the content.



MA.K12.MTR.5.1: Patterns and Structure
Students will be spending time focusing on relevant details in their pattern and creating plans to logically produce an overall image. This activity helps students recognize the patterns in the world around them and make connections to mathematical concepts.

1.H1.4 Wrap-Up: Cool Down

Component Overview: Students identify the transformations that show symmetry for two real-world objects.



1.H1.4 Wrap-Up: Cool Down

1. Symmetry is often found in nature. Describe the transformations that show the symmetry of each object.

Image	Object	Transformation
	Each petal	<i>Reflection and Rotation</i>
	Each arm of the starfish	<i>Reflection and Rotation</i>



Consider having the students take pictures of objects with symmetry and to share them with you for a class slideshow.

1.8 Line Segments

Lesson Introduction

	Benchmark	MA.912.GR.5.1 Construct a copy of a segment or an angle.			Learning Targets	- I can construct a copy of a line segment. - I can use the segment addition postulate.
	Benchmark Coherence	Building On			Working Toward	
		N/A			N/A	
	Essential Questions	- How can I make a copy of a line segment? - How are the lengths of pieces of a line segment related to the entire line segment? - How can I use the segment addition postulate to find the length of a line segment?				
	Lesson Narrative	Students are guided on how to make a copy of a line segment using a ruler, patty paper, compass, and toothpick. Using the ruler postulate, definition of collinear points, and the segment addition postulate students solve problems to determine the length of a line segment. Students work through the remainder of the lesson applying the segment addition postulate in mathematical and real-world applications.				
	Component Summary	1.8.1 Warm-Up (~5 min.)	Using critical thinking to find area (<i>SE p. 53</i>)	1.8.3 Guided Practice (10 min.)	Apply the segment addition postulate to solve mathematical problems (<i>SE p. 56</i>)	
		1.8.2 Guided Instruction (25 min.)	Copying line segments and developing the segment addition postulate (<i>SE p. 54</i>)	1.8.4 Try It! (5 min.)	Apply the segment addition postulate to solve a real-world problem (<i>SE p. 57</i>)	
		1.8.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 58</i>)			
	Resources	Glossary		Materials		Additional Preparation
		<u>Key Terms:</u> - Ruler Postulate - Collinear Points - Segment Addition Postulate <u>Additional Terms:</u> N/A - Line - Line Segment - Ray		- Ruler - Compass - Patty Paper - Toothpicks (Optional) Graph Paper		N/A

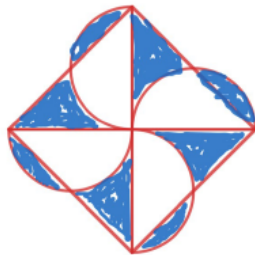
 Teacher Tips	Common Misconceptions	Things to Consider
	Students may swap the notation for a line segment and the length of the line segment by leaving off or including the segment symbol inappropriately.	- In 1.8.2 Guided Instruction, before introducing the Ruler Postulate, consider engaging in a quick number talk to help students recall how to find the distance between two points on a number line. Taking a moment here will help students to better understand the formal definition of the Ruler Postulate and answer question 5c in the same component. - If time is an issue, then consider assigning 1.8.4 Try It for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Compare and Connect In 1.8.2 Guided Instruction question 3, students are using a compass to copy a line segment. Students will choose where to place point M on their arc. Engage students in conversation of their placement of M , comparing their placements and connecting why all of their placements, as long as they are on the arc, are correct. Encourage the use of the appropriate vocabulary: circle, radius, center, and distance.	MA.K12.MTR.2.1: Multiple Representations Students build an understanding using constructions and express connections between the concepts and representations. In 1.8.2 Guided Instruction students copy line segments using a ruler, patty paper, compass, and toothpick. Students relate the measurement of a line segment using the Ruler Postulate and then apply the measurement using the Segment Addition Postulate.
	1.8.2 Guided Instruction question 4 provides students with a kinesthetic entry-point with the material. Using toothpicks, students can demonstrate a specific transformation. This activity will help students who need tactile stimulation digest the concepts.	
Lesson Notes:		

1.8.1 Warm-Up: Semicircles and Triangles

Component Overview: Students apply critical thinking to solve a question about the area of an image.



1.8.1 Warm-Up: Semicircles and Triangles



- What do you notice about the picture?
Answers will vary. Some is shaded, some is not. It has half circles.
- If the radius of each semicircle is 5, then what is the length of each of the square's diagonals?
20
- Dina determined the area of the shaded region using the drawing shown. Find the total shaded area.
 $A = 100 \text{ units}^2$



For students who are stuck remind them that the radius is the distance from the middle of the circle to the edge. This diagram tends to “trick” the eye by making it appear that the distance from the center of the square to the corner is the radius; however, this is the diameter. There are 4 iterations of the radius across the long diagonal.

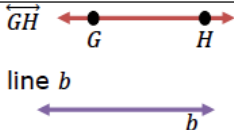
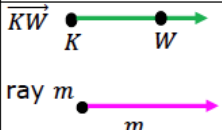
Allow students to struggle or to make mistakes. Depending on your class routine this may be a good problem for students to work on together. Limit the time students spend on the problem to about 5 minutes. Let students know that the expectation is to try and to think through the question and that it is okay to get it wrong.

1.8.2 Guided Instruction: Line Segments

Component Overview: Students review the terms *line*, *line segment*, and *ray* as well as how to name them. Next, students make a copy of a line segment using a ruler, patty paper, compass, and toothpick. The ruler postulate and collinear points are presented to develop the segment addition postulate. Students use the segment addition postulate to solve for a variable and for different segment lengths.



1.8.2 Guided Instruction: Line Segments

Line	Line Segment	Ray
A line is an undefined term.	A part of a line that is bounded by two distinct end points.	A part of a line that has a fixed starting point and extends infinitely in one direction.
		

When naming a line segment, the line segment symbol should be used; however, when giving the length of a line segment, the symbol should not be used.

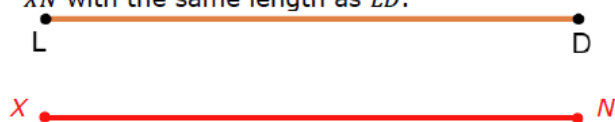
$$\overline{TM} \quad TM = 12 \text{ millimeters}$$

1. $\overline{RK}$ is shown. Use a ruler to measure the line segment in centimeters.



Note: Actual length will vary depending on student lesson format.

2. Use patty paper to copy $\overline{LD}$. Then, draw line segment $\overline{XN}$ with the same length as $\overline{LD}$.



Students may be familiar with these terms and representations. The formal introduction and correct naming of line, line segment, and ray is a fundamental part of geometry.

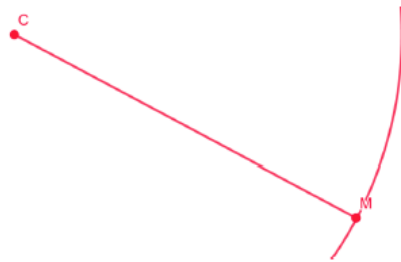


Before moving to question 1, consider asking the following:

- What are some ways that you can copy a line segment?
- How accurate are those ways?
- What tools can help copy a line segment?

1.8.2 Guided Instruction: Line Segments (continued)

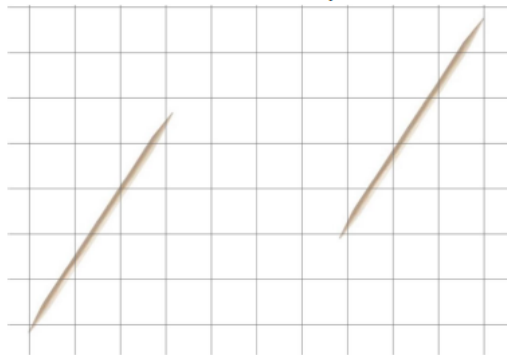
3. Follow these steps to draw a copy of $\overline{LD}$ using a compass.
 - Draw point C . This will be one endpoint of the new line segment.
 - Set the compass' point on point L .
 - Adjust the compass' width to point D so the compass' width is equal to the length of $\overline{LD}$.
 - Without changing the compass' width, place the compass' point on C and draw an arc roughly where the other endpoint will be.
 - Pick a point M on the arc that will be the other endpoint of the new line segment.
 - Draw line segment $\overline{CM}$ using a straightedge.



Engage students in the Literacy and Language Routine Compare and Connect to discuss the placement of point M . Encourage the correct use of vocabulary, including circle, radius, center, and distance.

1.8.2 Guided Instruction: Line Segments (continued)

4. Discuss with your partner how to show that the two toothpicks are congruent using each method in the table. Then, write a brief explanation for each method.



Transformation	<i>Translate each point of the left toothpick to the right seven units and up two units.</i>
Patty Paper	<i>Copy either toothpick onto the Patty Paper. Move the Patty Paper to cover the other toothpick.</i>
Compass	<i>Set the point of the compass on one end of a toothpick and open the compass to reach the other end of the same toothpick. Move the point on the compass to the end of the second toothpick and draw an arc. The toothpicks are congruent if the arc passes through the second endpoint of the second toothpick.</i>



Ruler Postulate

The points of a line can be placed in a one-to-one correspondence with the real numbers such that the distance between two distinct points is the absolute value of the difference of the corresponding real numbers.



Students may benefit from using toothpicks to explore transformations. For a kinesthetic approach, we are connecting using the compass and using a toothpick.



Students have been working with number lines since elementary school. Discuss that the concept of the ruler postulate is nothing new for them, just a formalized explanation.

1.8.2 Guided Instruction: Line Segments (continued)

5. $\overline{KC}$ and $\overline{YC}$ are shown.



- a. Discuss with your partner how aligning a ruler with $\overline{KC}$ illustrates the ruler postulate. Summarize your discussion.

Answers will vary. Place a ruler beside the line so that the points line up with positions on the ruler. Subtract the corresponding number on the ruler for one point from the corresponding number on the ruler for the other point. The absolute value of this difference is the length of the segment between the two points.

- b. Use a ruler to measure each line segment. Include units.

$KY = 4.4 \text{ cm}$ $YC = 6.5 \text{ cm}$ $KC = 10.9 \text{ cm}$
Actual length may vary based on units of measure.

- c. Discuss with your partner why $KY + YC = KC$ should be true.

Answers will vary. The line is equal to the sum of its parts.

Points on the same line are called **collinear points**.



Segment Addition Postulate

If points A , X , and B are collinear points and point X is between points A and B , then $AX + XB = AB$.

6. $\overline{KW}$ and $\overline{WT}$ are shown, where $KW = 4x - 7$, $WT = 23.5$, and $KT = 7x - 3$.



- a. Determine the value of x .

$x = 6.5$

- b. What is the length of $\overline{KW}$?

$KW = 19$



To provide relevance to the students, lead a discussion of where the segment addition postulate may be useful in the real world. Consider a variety of professions that may use line segments, like architecture or graphic design. Consider manipulatives, like Lego and K'nex as places they may have used line segment addition in the past.



Monitor students as they solve the equation in question 6. Note students who appear struggling. Consider utilizing a small group during 1.8.3 Guided Practice.

1.8.3 Guided Practice: Segment Addition

Component Overview: Students will apply the segment addition postulate and solve for variables and segment lengths.



1.8.3 Guided Practice: Segment Addition

1. $\overline{YB}$, $\overline{BW}$, and $\overline{WH}$ are shown, where $YB = 2x$, $BW = x + 4$, $WH = 3x - 1$, and $BH = 35$.



- Determine the value of x .
 $x = 8$
 - What is the length of $\overline{YW}$?
 $YW = 28$
2. Points K , L , M , and N are collinear such that $KM = LN$.

- Draw a diagram that represents points K , L , M , and N .

Answers will vary.



- Compare your diagram with your partner. Discuss whether or not the diagrams drawn must all be the same. Summarize your discussion.

Answers will vary. No, they do not have to be the same. As long as all four points are collinear and $KM = LN$, they could be placed anywhere along the line.

- Determine the value of x if $KM = 6x - 6.2$ and $LN = 5x - 2$.
 $x = 4.2$
- Using the information from c, determine the length of $\overline{KM}$.
 $KM = 19$



Continue to monitor students as they solve equations. Consider directing students to the Student Area online for a refresher on solving equations.



Question 2 leaves students to test out the placement of points when not given an image and may try labeling the points in a variety of ways. Because of the declaration that $\overline{KM} = \overline{LN}$, then the equations are to be set equal, and the answer for the variable will not change.

1.8.4 Try It: Ocean to Lake Florida Trail

Component Overview: Students will use the segment addition postulate to solve a real-world problem.



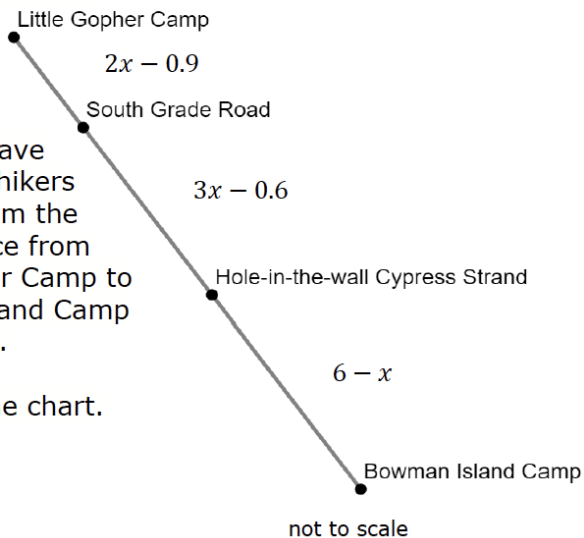
1.8.4 Try It: Ocean to Lake Florida Trail

- The Ocean to Lake Hiking Trail runs for approximately 63 miles, west from Hobe Sound Beach to Lake Okeechobee. Leigh Ann created a simple map of part of the trail that goes through J.W. Corbett Wildlife Management Area.

Her map is shown.

Leigh Ann gave the map to hikers and told them the total distance from Little Gopher Camp to Bowman Island Camp is 10.5 miles.

Complete the chart.



Consider showing students a map of the Ocean to Lake hiking trail and discuss why a simple map may be useful.



Students can assess and evaluate the reasonableness of their solutions, MA.K12.MTR.6.1, by substituting their answers in for all lengths in the problem. If the smaller line segment lengths sum to equal the entire length, then students can feel confident that their answer is correct.

Hike	Distance (miles)
Little Gopher Camp to South Grade Road	2.1 miles
South Grade Road to Hole-in-the-wall Cypress Strand	3.9 miles
Hole-in-the-wall Cypress Strand to Bowman Island Camp	4.5 miles

1.8.5 Wrap Up: Cool Down

Component Overview: *Students create an example demonstrating the Segment Addition Postulate.*



1.8.5 Wrap-Up: Cool Down

1. Create an example of the Segment Addition Postulate and show the steps to finding the total length.

Answers will vary.

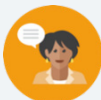


Student responses can be used to identify misconceptions and plan for upcoming instruction.

1.9 Copying an Angle

Lesson Introduction

 Benchmark	MA.912.GR.5.1 Construct a copy of a segment or an angle.		 Learning Targets	- I can construct a copy of an angle. - I can use the angle addition postulate.	
 Benchmark Coherence	Building On			Working Toward	
	N/A			N/A	
 Essential Questions	- What are the steps to construct the copy of an angle? - How can I use the angle addition postulate to determine the value of unknown variables and angle measures?				
 Lesson Narrative	Students work together to determine the order of the steps needed to construct the copy of an angle. After deciding on an order, they will test the steps, make adjustments as needed, and test again.				
 Component Summary	1.9.1 Warm-Up (~5 min.)	Bell Work using the segment addition postulate (<i>SE p. 59</i>)	1.9.3 Guided Instruction (10 min.)	Angle addition postulate (<i>SE p. 61</i>)	
	1.9.2 Activity (20 min.)	Determining the steps to construct the copy of an angle (<i>SE p. 59</i>)	1.9.4 Try It (10 min.)	Constructing the copy of an angle (<i>SE p. 62</i>)	
	1.9.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 63</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> Angle Addition Postulate <u>Additional Terms:</u> N/A		- Compass - Protractor - Patty Paper (Optional) Chart Paper		Make copies of the steps and cut into strips for students to use when determining the order of the steps for constructing the copy of an angle in 1.9.2 Activity.

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may think that the length of the rays determines whether an angle is a copy of another.	If a document camera or Smart Board are not available, consider making an oversized copy of the steps for constructing a copy of an angle. These steps can be used when providing the final answer and discussion of the step order at the end of 1.9.2 Activity.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Students should work in pairs to order the steps in 1.9.2 Activity. After thinking about the order individually, students can share their progress with their partner. Partner pairs then find another partner pair to share their thinking with. Providing this structure gives students an opportunity to feel confident about their order before testing if it is correct.	MA.K12.MTR.1.1: Participate in Effortful Learning Students are building perseverance by testing and modifying their steps. Students should be encouraged to stay engaged and maintain a positive mindset, helping and supporting each other through 1.9.2 Activity. Using logic to order the steps is developing skills and a foundation for writing proofs.
 Differentiate Instruction	For 1.9.2 Activity, consider providing students with the correct placement of more of the steps in the beginning so that they have fewer steps to order. For example, provide the correct placement of steps 1, 2, 3, 6, and 9.	
Lesson Notes:		

1.9.1 Warm-Up: Bell Work

Component Overview: Students will use the segment addition postulate to solve for an unknown value and the length of a segment.



1.9.1 Warm-Up: Bell Work

1. Points C , H , and W are collinear. Point W is between C and H . $CW = 4x - 2$, $HW = 5x - 19$, and $CH = 7x + 5$.
 - a. Find the value of x .
 $x = 13$
 - b. What is the length of $\overline{HW}$?
 $HW = 46$



Use student understanding to inform your instruction and to determine where small group instruction may be helpful during this and further lessons. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.

1.9.2 Activity: Copying an Angle

Component Overview: Students collaborate on ordering the steps for constructing a copy an angle. After sharing their order with others, the students will follow the steps to copy an angle. Make sure students have adequate time to attempt, fail, and adjust the steps if needed.

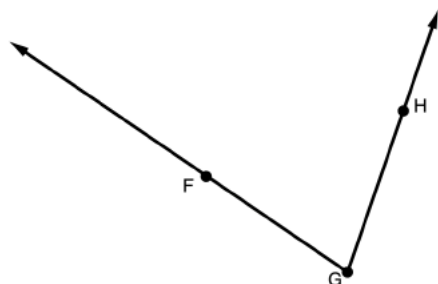


1.9.2 Activity: Copying an Angle

1. $\angle FGH$ is shown.

- a. Discuss with your partner how to copy $\angle FGH$ using only a compass.

- b. Work with your partner to order the nine steps for copying $\angle FGH$. The first and last step are in the correct position.



The intention of this activity is to have students collaborate to order the steps for constructing the copy of an angle. Consider making copies of the provided steps and cut them into strips. Have the students put the strips in the order they think will provide the steps for constructing the copy of an angle. Student partners should compare their ordering with other partner groups. To verify if their ordering is correct, students should follow the steps to construct the copy of an angle. If the steps are not in the correct order, students can adjust to correct themselves and verify again. Once the entire activity is completed, students can number the steps in the correct order in the table below for future reference.

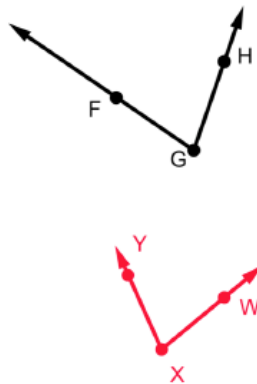
Students learn more by making mistakes. Provide students with adequate time to order the steps, test the steps, and adjust the steps to try again.

MA.K12.MTR.1.1: Participating in Effortful Learning
Students are building perseverance by testing and modifying their steps. Students should be encouraged to stay engaged and maintain a positive mindset, helping and supporting each other through the activity. Using logic to order the steps is developing skills and a foundation for writing proofs.

1	Draw a point and label it X . Then, draw a ray starting at point X in any direction. Label the ray $\overrightarrow{XZ}$.
7	Set the compass' point on point A and adjust the compass' width to point B .
8	Without changing the compass' width, place the compass' point on Y and draw a similar arc to create point W where the two arcs meet.
2	Set the compass' point on point G and adjust the compass' width to any width.
6	Without changing the compass' width, place the compass' point on X and draw a similar arc. Label the point where the arc and $\overrightarrow{XZ}$ intersect point Y .
4 or 5	Label the point where the arc crosses $\overrightarrow{GF}$ as point A .
4 or 5	Label the point where the arc crosses $\overrightarrow{GH}$ as point B .
3	Draw an arc across both rays of $\angle FGH$.
9	Draw $\overrightarrow{XW}$ using a straightedge.

1.9.2 Activity: Copying an Angle (continued)

- c. Compare your ordering with another group. If necessary, make adjustments to the order.
Answers will vary.
- d. Use your steps to draw $\angle WXY$, a copy of $\angle FGH$. If necessary, make adjustments to the order of your steps.



To review the correct ordering, select a group to share their ordering with the class. This can be done with a document camera, a Smart Board, or by making large copies of the steps that can be placed in order on the board. If a classmate doesn't agree with the ordering, the classmate should question the presenters and provide reasoning for why the order is incorrect. This continues until the class agrees upon the correct order.

Consider allowing students to edit the steps. Some students may consider the process easier if some steps are combined. Help students understand that memorizing the steps is not the point. This can be related to common tasks students do that require steps, but they do not have the steps memorized. Tying one's shoes is such a task. Students know the process but do not need to have the steps memorized. Alternately, they could likely figure out if steps are correct based on their experience.

1.9.3 Guided Instruction: Angle Addition Postulate

Component Overview: Students discover the angle addition postulate.



1.9.3 Guided Instruction: Angle Addition Postulate



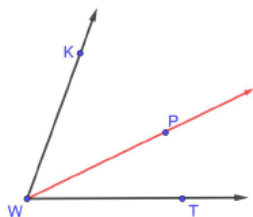
Angle Addition Postulate

If point X is in the interior of $\angle AMB$, then
 $m\angle AMX + m\angle XMB = m\angle AMB$.



The Angle Addition Postulate typically makes sense to students. Use 1.9.3 Guided Instruction as an opportunity to remind students the difference between notations such as $\angle ABC$ and $m\angle ABC$.

- Point P lies in the interior of $\angle KWT$ as shown.



- Draw $\overline{WP}$.
Shown on diagram.
- Use a protractor to measure each angle in degrees.
 $m\angle KWT = 70^\circ$ $m\angle KWP = 44^\circ$ $m\angle PWT = 26^\circ$
- Discuss with your partner why
 $m\angle KWP + m\angle PWT = m\angle KWT$ should be true.
 Summarize your discussion.
The two smaller angles should add up to the measurement of the large angle.
- If $m\angle KWT = 70^\circ$, $m\angle KWP = (5x + 5)^\circ$ and
 $m\angle PWT = (4x - 7)^\circ$, determine the value of x .
 $x = 8$

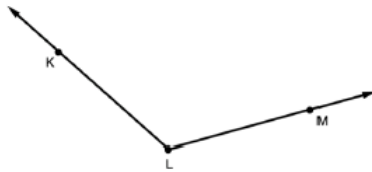
1.9.4 Try It: Copying an Angle

Component Overview: Students construct a copy of a given angle.



1.9.4 Try It: Copying an Angle

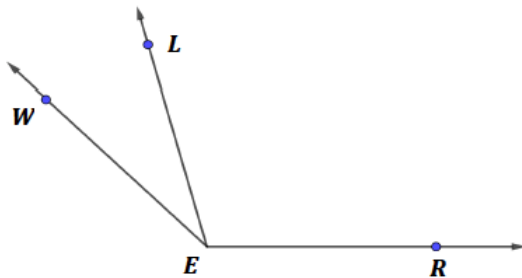
1. $\angle KLM$ is shown.



Draw $\angle RST$, a copy of $\angle KLM$.



2. Three angles are shown, $\angle WEL$, $\angle LER$, and $\angle WER$.



If $m\angle LER = (14.35x - 9.8)^\circ$, $m\angle WEL = (2.5x + 10)^\circ$, and $m\angle WER = 135^\circ$, what is $m\angle WEL$?

$m\angle WEL = 30^\circ$



Monitor students to make sure the appropriate construction marks are placed to ensure the correct construction. Also, monitor that students are labeling the points on the copy, as this is a necessary part of the process. Students can use patty paper or a protractor to assess the accuracy of the copy.

1.9.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



1.9.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about copying an angle.

Answers will vary.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

1.10 Segment Bisectors

Lesson Introduction

 Benchmark	MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.		 Learning Targets	- I can construct the bisector of a segment. - I use the definition of a midpoint to solve problems.	
 Benchmark Coherence	Building On			Working Toward	
	N/A			N/A	
 Essential Questions	- Where is a segment bisector located? - How can the midpoint be determined? - How can knowing midpoints help solve for variables?				
 Lesson Narrative	Through exploration with the use of patty paper, students determine the relationship between the midpoint of a segment and a segment bisector while also making connections to transformations. Using this relationship, students will solve for unknown variables and find the length of line segment.				
 Component Summary	1.10.1 Warm-Up (~5 min.)	Notice and wonder about longest pedestrian bridge (<i>SE p. 64</i>)	1.10.4 Guided Practice (10 min.)	Practice using segment bisectors to solve for segment lengths (<i>SE p. 67</i>)	
	1.10.2 Exploration (15 min.)	Exploring midpoint and segment bisectors (<i>SE p. 64</i>)	1.10.5 Your Turn! (5 min.)	Independent practice using segment bisectors to solve for segment lengths (<i>SE p. 68</i>)	
	1.10.3 Guided Instruction (10 min.)	Discover using segment bisectors to solve for segment lengths (<i>SE p. 66</i>)	1.10.6 Wrap-Up (~5 min.)	Error Analysis of solving for segment lengths (<i>SE p. 68</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Midpoint - Segment Bisector <u>Additional Terms:</u> N/A		- Patty Paper - Ruler (Optional) Geoboards (Optional) Tape or Glue		1.10.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider	
	- Students may think a segment bisector must intersect at a right angle. - Students may struggle setting up the equations for solving for the line segment lengths.	- Always mark the given information on the diagram so that students can see the relationships between the lengths of the different line segments. - Consider using information from previous lessons to determine which students could use help in solving equations. Work with these students during 1.10.4 Guided Practice and 1.10.5 Your Turn! while releasing students who can solve equations to work individually. - If time is an issue, consider assigning 1.10.5 Your Turn! for homework.	
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard	
	Notice and Wonder Students will explain what they notice about the image of the bridge provided in 1.10.1 Warm-Up. This routine provides a safe place for students to brainstorm and to consider objects without any fear of being wrong. Encourage students to notice beyond the obvious. Instill in your students the idea of wonder. Consider helping students frame their wondering by asking questions, such as: - What do you think an 8-year-old would wonder? - What do you think your parent/guardian would wonder? - What do you think I would wonder? Removing the pressure of being right may help free students to imagine.	MA.K12.MTR.5.1: Patterns and Structure Throughout the lesson, students will use details from given information to solve for segment lengths. By connecting the concepts of midpoint and bisector, the students will use and mark images to set up the relationships needed to write the equations used for determining the segment lengths.	
 Differentiate Instruction	For 1.10.1 Warm-Up question 2, using Geoboards, give students an opportunity to explore other shapes they can make with overlapping straight lines. This kinesthetic activity may encourage students to become engaged with the course. An alternative to a Geoboard could be a square piece of wood with posts lining the edges. Students can tie strings to the posts and recreate this same image.		
Lesson Notes:			

1.10.1 Warm-Up: Notice and Wonder

Component Overview: Students begin this lesson by looking at an image of the longest pedestrian footbridge. Students state what they notice and wonder.



1.10.1 Warm-Up: Notice and Wonder

In May 2021, the world's longest suspended pedestrian bridge opened in Arouca, Portugal. The bridge runs over 1700 feet across a canyon, nearly 600 feet above the Paiva River.



1. What do you notice?

Answers will vary. I notice that there are a lot of geometric figures used to create the bridge, including hexagons and triangles. I notice the bridge sags in the middle.

2. What do you wonder?

Answers will vary. I wonder if the bridge sways back and forth in the wind. I wonder how long it took to build the bridge. I wonder how they were able to construct the bridge over the canyon.



Notice and Wonder

Students will have many things they notice. Encourage students to take risks and let their creativity explore options.



Students who are particularly intrigued by the bridge could further explore the engineering of the bridge.

1.10.2 Exploration: Segment Bisector

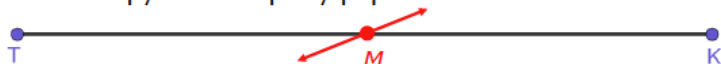
Component Overview: Using patty paper, students will identify the midpoint and draw a segment bisector. Through discussion, students will determine which transformations will map half of a line segment onto another. Using the definition, students draw different types of segment bisectors and the point of intersection is defined as the midpoint.



1.10.2 Exploration: Segment Bisector

1. Complete the following.

a. Copy $\overline{TK}$ onto patty paper.



b. Follow the steps on the copy of $\overline{TK}$.

- Fold the copy of $\overline{TK}$ that is on patty paper so the endpoints T and K align.
- Place point M on $\overline{TK}$ where the paper folds.
- Copy point M onto $\overline{TK}$ above.
- Draw a line, a line segment, or a ray through point M .

c. Discuss with your partner what you notice about $\overline{TM}$ and $\overline{MK}$.

Answers will vary. They are the same length.

The **midpoint** of a segment divides it into two congruent segments.

2. $\overline{WX}$ is shown where point P is the midpoint.



a. Discuss with your partner what transformations can be used to show that $\overline{WP}$ maps to $\overline{PX}$.

Answers will vary. Transformations that can show that $\overline{WP}$ maps to $\overline{PX}$ are

- a rotation 180° at point P .*
- a reflection over point P .*
- a translation of $\overline{WP}$ to the right.*



Students may or may not draw the segment bisector intersecting the line segment at a right angle. As long as it is drawn through the midpoint, any angle with the line segment is acceptable.

Consider allowing students to tape or paste their patty paper into their workbook.



This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt in question 2a and what students share in question 2b to determine student thinking to highlight or misconceptions to address.

1.10.2 Exploration: Segment Bisector (continued)

- b. Ask two classmates which transformations they selected. Record their transformation and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

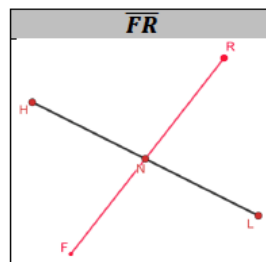
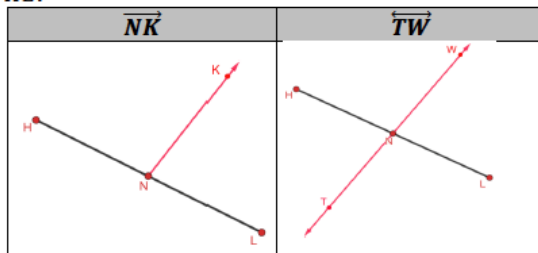
Answers will vary.

Transformation	My Summary of Their Reason	Initials

A **segment bisector** is a line, ray, or line segment that cuts a line segment into two equal parts.

3. $\overline{HL}$ is shown where point N is the midpoint.

- a. Use each description to draw a segment bisector for $\overline{HL}$.



Having students listen to their classmates and then write down what the classmate said helps the student think through their own response. Students may hear different ways to express the same thing, thus building their own mathematical thinking.



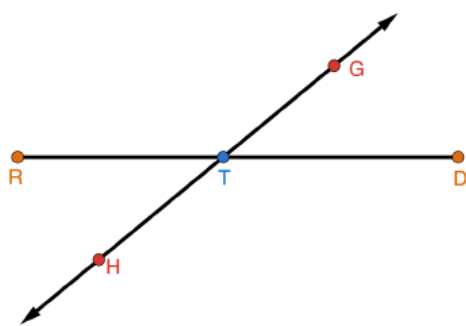
Segment bisectors are not required to intersect the line segment at right angles. Any angle is acceptable.

1.10.2 Exploration: Segment Bisector (continued)

- b. Compare your drawings with your partner.
- c. Using the definition of a segment bisector, discuss with your partner if all of the segment bisectors have to be at the same angle with $\overline{HL}$. Summarize your discussion.

Answers will vary. No, a bisector only has to cross in the middle, the angle it crosses is not important.

4. $\overline{RD}$ and $\overline{HG}$ are shown where point T is the point of intersection.



- a. Use patty paper to copy $\overline{RD}$.
- b. Fold the copy of $\overline{RD}$ to determine if point T is a midpoint.
- c. Complete the sentence.

$\overline{HG}$ ☐ is ☒ is not a segment bisector because point T

☐ is ☒ is not a midpoint.



Students may incorrectly fold their paper along line HG instead of folding at point T so that $\overline{RT}$ coincides with $\overline{TD}$. If you notice this happening, instead of correcting them, ask questions about which line they are trying to bisect, which segments they are trying to determine, and if they have equal lengths.

1.10.3 Guided Instruction: Determining Lengths

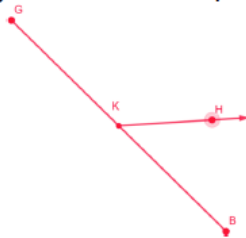
Component Overview: Using the definition of a segment bisector, solve for unknown variables and line segment lengths.



1.10.3 Guided Instruction: Determining Lengths

1. $\overrightarrow{KH}$ bisects $\overline{GB}$ at point K .

- a. Draw a diagram of the description.

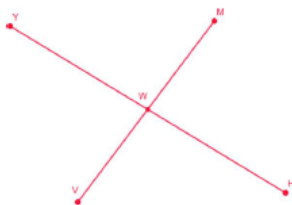


- b. Determine the value of x if $GK = 7x + 36$ and $KB = 13x - 18$.
 $x = 9$

- c. What is the length of $\overline{GB}$?
 $GB = 198$

2. $\overline{YH}$ and $\overline{MV}$ bisect each other at point W .

- a. Draw a diagram of the description.



- b. Determine the value of b if $YH = 12b - 4.5$, $MV = 17b + 0.75$, and $MW - WH = 13.25$.
 $b = 4.25$



Take care to ensure that students understand that the angle at which ray KH intersects line segment GB does not have to be 90 degrees. In fact, the preciseness of creating such an angle requires its own construction which will be addressed in unit 2.



While demonstrating question 2b, mark the diagram with the lengths that are given and can be determined from the problem. While doing this, consider asking the following questions:

- How is the length of $\overline{YW}$ related to the length of $\overline{YH}$? How can I represent this algebraically?
- How are the lengths of $\overline{YW}$ and $\overline{WH}$ related?
- How is the length of $\overline{VW}$ related to the length of $\overline{VM}$? How can I represent this algebraically?
- How are the lengths of $\overline{VW}$ and $\overline{WM}$ related?

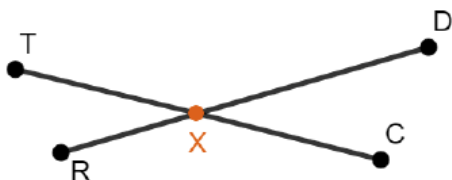
1.10.4 Guided Practice: Determining Lengths

Component Overview: Using the definition of a segment bisector, solve for unknown variables and line segment lengths.



1.10.4 Guided Practice: Determining Lengths

1. $\overline{RD}$ bisects $\overline{TC}$ at point X .



- a. Determine the value of x if $TX = \frac{16}{3}x - 41$ and $XC = 3x + 22$.
 $x = 27$
- b. What is the length of $\overline{TC}$?
 $TC = 206$



Consider pairing students together for this question and naming them Partner 1 and Partner 2. For question 1a, have Partner 1 do all of the talking while Partner 2 does not talk, and only writes what Partner 1 says. When Partner 1 is finished talking, Partner 2 can agree with the work or highlight in the work where an error occurred.

For question 1b, students swap roles.

1.10.5 Your Turn!

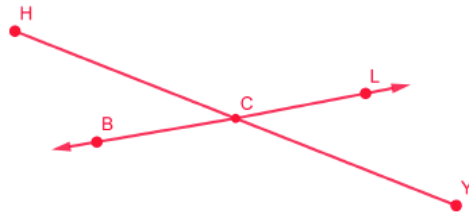
Component Overview: Students draw a diagram of the description, solve for the variable, and determine the length of a line segment.



1.10.5 Your Turn!

1. $\overleftrightarrow{BL}$ bisects $\overline{HY}$ at point C .

- a. Draw a diagram of the description.



- b. Determine the value of x if $HC = 5x + 9$ and $CY = 8x - 36$.
 $x = 15$

- c. What is the length of $\overline{HY}$?
 $HY = 168$



Consider asking these follow-up questions:

- Why is it important that we are asked to find the midpoint of the line segment rather than the line?
- Why might it be impossible to find the midpoint, or draw a bisector of a line, but it is possible on a line segment?

1.10.6 Wrap-Up: Error Analysis

Component Overview: Students will read through the work of other students and write notes to the students indicating their mistakes.

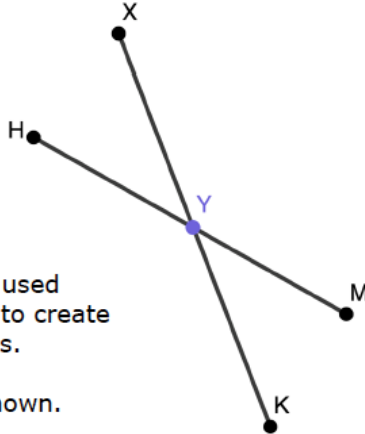


1.10.6 Wrap-Up: Error Analysis

- $\overline{XK}$ and $\overline{HM}$ bisect each other at point Y , where $XY = 5x + 3$, $HY = 2y + 5$, $XK = 11x + 2$, and $YM = 3y - 2$.

Taliyah and Giancarlo used the given information to create two different equations.

Their equations are shown.



This question assesses student understanding of using segment bisectors to solve problems. Students' responses to this activity can provide data on which students may need additional support with this content.

Taliyah	Giancarlo
$5x + 3 = 2y + 5$	$5x + 3 = 2(11x + 2)$

Both Taliyah and Giancarlo made a mistake. If Taliyah was trying to find the value of y and Giancarlo was trying to find the value of x , determine each of their errors. Explain each error.

Taliyah	Giancarlo
Taliyah set the wrong line segments equal to each other. She should have set $HY = YM$ to create the equation $2y + 5 = 3y - 2$.	Giancarlo doubled the length of the full line segment, $\overline{XK}$, when he should have taken half the length and set equal to XY .

1.11 Perpendicular Bisectors

Lesson Introduction

 Benchmark	MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.	 Learning Targets	- I can construct a perpendicular bisector. - I can use rigid transformations to show that any triangle formed by drawing two segments from the endpoints of a segment to a point on its perpendicular bisector is isosceles. - I can use the properties of a perpendicular bisector to solve problems.	
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What key features define a perpendicular bisector? - How can a perpendicular bisector help in constructing an isosceles triangle? - If a perpendicular bisector exists in a triangle, which side lengths must be congruent?			
 Lesson Narrative	Students construct and examine perpendicular bisectors, determining that points on a perpendicular bisector are of equal distance to the endpoints and form an isosceles triangle. Students use the relationship between the line segment and the perpendicular bisector to solve problems.			
 Component Summary	1.11.1 Warm-Up (~5 min.)	Comparing the areas of three inscribed figures (<i>SE p. 69</i>)	1.11.4 Bring It Together (5-10 min.)	Relationships formed between a line segment and a perpendicular bisector (<i>SE p. 73</i>)
	1.11.2 Exploration (10 min.)	Constructing and examining points on a perpendicular bisector (<i>SE p. 70</i>)	1.11.5 Guided Practice (5 min.)	Solving using a perpendicular bisector (<i>SE p. 73</i>)
	1.11.3 Activity (10 min.)	Creating an isosceles triangle from a perpendicular bisector (<i>SE p. 72</i>)	1.11.6 Your Turn! (5 min.)	Practice with perpendicular bisectors (<i>SE p. 74</i>)
	1.11.7 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 74</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Right Triangle - Isosceles Triangle <u>Additional Terms:</u> - Perpendicular Bisector		- Compass - Ruler - Patty Paper (Optional) String (Optional) Highlighters	
			Additional Preparation Consider preparing sentence stems for question 6 in 1.11.2 Exploration.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not use a wide enough compass width to form intersecting arcs. - Students may change the width of their compass when measuring from each endpoint.	- If you are in an environment where students can go outside, consider scaling up these activities by replacing a compass with long string tied around a tree or post. - Consider using the GeoGebra applet for constructing a perpendicular bisector in 1.11.2 Exploration. - If time is an issue, then consider assigning 1.11.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Number Talk When students begin 1.11.5 Guided Practice, a Number Talk would be a great way to get students discussing and working through these questions. This strategy should work well here because of the different ways for students to work through each equation.	MA.K12.MTR.2.1: Multiple Representations Students will build and demonstrate understanding by constructing a perpendicular bisector to represent equal distance to the endpoints of a line segment in 1.11.2 Exploration. Using manipulatives helps students make connections between concepts and representations.
 Differentiate Instruction	For the vocabulary in this section, have students identify the key words of the definition by highlighting or underlining each word that is critical to the definition. Students should then define or explain each of the key words. Once identified, have students read to each other the vocabulary word and the keys words. - Identify key words - Eliminate non-critical key words - Define each word - Share with partner	
Lesson Notes:		

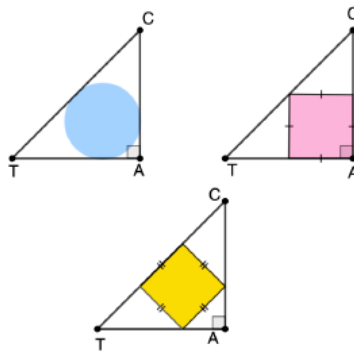
1.11.1 Warm-Up: Compare Areas

Component Overview: Students will justify which shaded region has the greatest area.



1.11.1 Warm-Up: Compare Areas

$\triangle CAT$ is an isosceles right triangle.



Students should be free to make their own determination and justification. Consider not requiring students to use precise language in their justification. If time, have students share their choice with a partner and to verbalize their justification.

- Which inscribed figure has the greatest area? Justify your choice.

Answers will vary. I think the triangle with a circle in it will have the largest area because the curved parts can cover more than a straight edge.

1.11.2 Exploration: Constructing Perpendicular Bisectors

Component Overview: Students construct the perpendicular bisector of a line segment. After measuring the distance from various points, students write a conjecture about the distance of any point on the perpendicular bisector to the endpoints of the line segment.



1.11.2 Exploration: Constructing Perpendicular Bisectors

1. Use the steps shown to construct the perpendicular bisector of $\overline{AB}$.

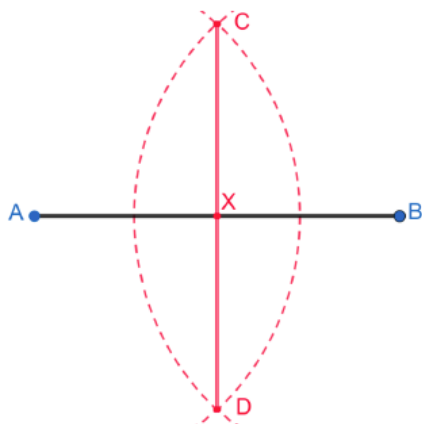
Step 1: Set the compass point at point A and open the compass to a width approximately two-thirds the length of $\overline{AB}$.

Step 2: Without changing the width of the compass, draw an arc above and below the line centered at point A .

Step 3: Repeat Steps 1 and 2 to create an arc centered at point B .

Step 4: Mark the two points of intersections of the arcs as C and D . Draw a line creating $\overline{CD}$ that intersects $\overline{AB}$.

Step 5: Label the point of intersection for $\overline{CD}$ and $\overline{AB}$ as point X .



Consider using the GeoGebra applet for 1.11.2 Exploration.



1.11.2 Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize in 1.11.4 Bring It Together. Students may revise their thinking throughout the exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group during 1.11.4 Bring It Together.



Students typically struggle placing arcs that intersect. As you circulate, ensure that students are using compass widths that are greater than half the length of the line segment. Students may need to be reminded to keep their compass setting the same when locating point C and also the same when locating point D .

1.11.2 Exploration: Constructing Perpendicular Bisectors (continued)

2. Use a protractor and ruler to find each of the measurements.

Answers will vary depending on students' compass placement. The measures of segments AX and BX are half of the measurement found for segment AB . $CB = CA$ and $DB = DA$.

Angle	Measurement	Segment	Measurement
$\angle AXD$	90°	$\overline{AB}$	6.2 cm
		$\overline{AX}$	3.1 cm
$\angle AXC$	90°	$\overline{BX}$	Answers will vary.
		$\overline{CB}$	Answers will vary.
$\angle BXD$	90°	$\overline{DB}$	Answers will vary.
		$\overline{CA}$	Answers will vary.
$\angle BXC$	90°	$\overline{DA}$	Answers will vary.



Students can use any unit to measure the line segments but must use the same unit of measure for all of them.

3. Complete the statements.

The angle formed by the line of intersection of $\overline{AB}$ and $\overline{CD}$ is

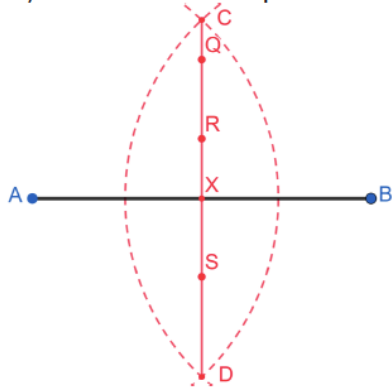
- ☐ 85°
☒ 90°
☐ 105°

The perpendicular bisector $\overline{CD}$ divides

- ☒ 2
☐ 3
☐ 4

$\overline{AB}$ into _____ congruent parts.

4. On the perpendicular bisector $\overline{CD}$, add three points labeled Q , R , and S in different places.



1.11.2 Exploration: Constructing Perpendicular Bisectors (continued)

5. Using a ruler, measure the length of each line segment.

Answers will vary depending on the placement of the points along $\overline{CD}$.

Segment	Length
$\overline{QA}$	
$\overline{QB}$	
$\overline{RA}$	
$\overline{RB}$	
$\overline{SA}$	
$\overline{SB}$	

6. Write a conjecture about the endpoints of a line segment and any point on its perpendicular bisector.

Answers will vary. The distance from an end point to any point on the perpendicular bisector will be the same length as the other end point of the same line and that point on the perpendicular bisector.

7. Share your conjecture with a partner. Summarize your partner's conjecture and have them initial that your summary is correct.

Answers will vary.

Partner's Conjecture	Partner's Initials



Creating a conjecture can be intimidating for students. Consider providing students with sentence stems to help scaffold their thinking.

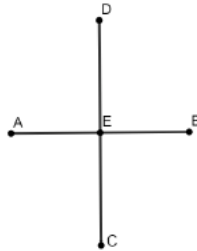
1.11.3 Activity: Perpendicular Bisectors and Triangles

Component Overview: Students use patty paper and transformations to determine the definition of an isosceles triangle.

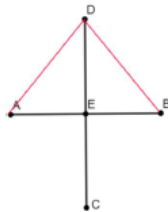


1.11.3 Activity: Perpendicular Bisectors and Triangles

- Copy $\overline{AB}$ and its perpendicular bisector $\overline{CD}$ on patty paper.



- Draw $\overline{AD}$ and $\overline{BD}$.

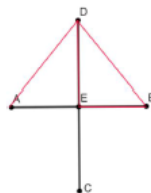


- Make a conjecture about $\triangle ADE$ and $\triangle BDE$.
Answers will vary. Both triangles are right triangles. The lines drawn to make the triangles are both the same lengths.



A **right triangle** is a triangle containing an interior right angle.

- Reflect $\triangle ADE$ over $\overline{CD}$.



1.11.3 Activity supports vocabulary terms that students should already be familiar with while also foreshadowing SAS congruence of triangles. The congruence of the triangles is shown using transformations which spirals the content previously learned in Unit 1.

1.11.3 Activity: Perpendicular Bisectors and Triangles (continued)

5. Discuss with your partner what the reflection shows is true about $\triangle ADE$ and $\triangle BDE$. Summarize your discussion.
The triangles are congruent.



An **isosceles triangle** contains at least two equal length sides and two equal interior angle measures.

6. Complete the statements.

In $\triangle ABD$, $\angle DAB$ is congruent to ☐ $\angle ADB$
☐ $\angle DBA$ because a

reflection is a rigid motion that preserves angle measure.

Since $m\angle DAB$ is equal to ☐ $m\angle ADB$,
☐ $m\angle DBA$, and ☐ $AD = AB$,
☐ $AD = BD$,
☐ $AE = BE$,
☐ None of these

by definition $\triangle ABD$ is a(n) ☐ scalene
☐ **isosceles**
☐ equilateral triangle.



Consider asking one of the partner pairs to read the completed statements to the as a way to verify their choices. This will provide an auditory entry-point for learners from listening.

1.11.4 Bring It Together: Perpendicular Bisectors and Triangles

Component Overview: Students develop and test statements about points on a perpendicular bisector. They then use a ruler to confirm the definitions.



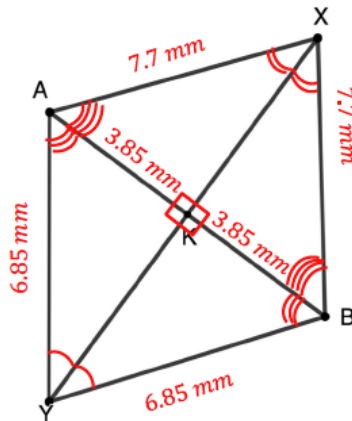
1.11.4 Bring It Together: Perpendicular Bisectors and Triangles



If a point is on the perpendicular bisector of a line segment, then it is equidistant from the endpoints of the line segment.

In the diagram, $\overline{XY}$ is the perpendicular bisector of $\overline{AB}$, $XA=7.7$ millimeters (mm), $YB=6.85$ mm, and $AB=7.7$ mm. The figure is not drawn to scale.

1. Label all segment measurements from the given information on the image. Then, label all other segments based on the relationship of a line segment and its perpendicular bisector.



2. Given $\overline{XY}$ is the perpendicular bisector of $\overline{AB}$, label all of the congruent angles.



The triangle formed by a line segment and two segments drawn from the endpoints of the segment to a point on the perpendicular bisector of the segment is an isosceles triangle.



Consider debriefing any questions that students had in 1.11.2 Exploration and 1.11.3 Activity. Also, consider highlighting student voices to share rich discussions or comments overhead while they collaborated for the whole class to hear or to address common misconceptions that were evident among the majority of the class.



After students have labeled the segments from the given information, consider asking the following questions:

- Is the length of $\overline{YK}$ equal to the length of $\overline{XK}$? How do you know?
- How could you determine the lengths of $\overline{YK}$ and $\overline{XK}$?

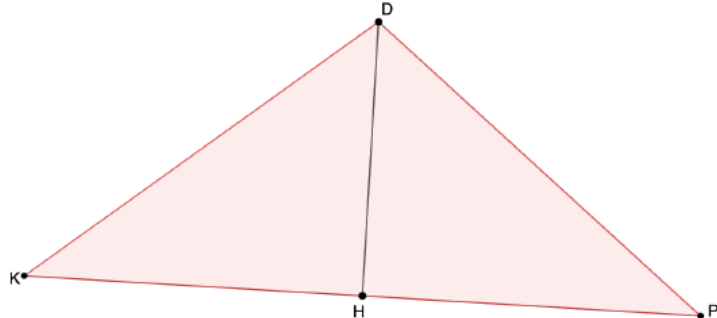
1.11.5 Guided Practice: Perpendicular Bisectors

Component Overview: Students use a perpendicular bisector to solve for an unknown variable and a segment length.



1.11.5 Guided Practice: Perpendicular Bisectors

1. $\overline{DH}$ is the perpendicular bisector of $\overline{KP}$, where $KD = -x - 1$ and $PD = -\frac{x}{2} + 19$.



- What is the value of x ?
 $x = -40$
- What is the length of $\overline{KD}$?
 $KD = 39$



Allow informal observation throughout the previous two components to dictate how “guided” 1.11.5 Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

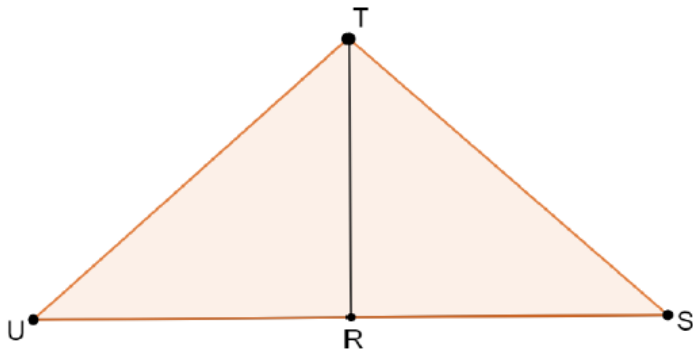
1.11.6 Your Turn!

Component Overview: Students use the perpendicular bisector to solve for an unknown variable. Then, a construction with an error is shown, where students describe the error.



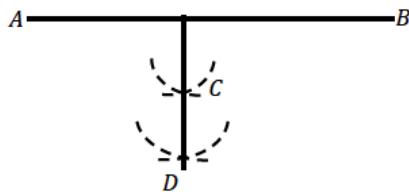
1.11.6 Your Turn!

- $\overline{RT}$ is the perpendicular bisector of $\overline{US}$. $UT = 8y - 12$ and $ST = 3y + 23$. Find the value of y .
 $y = 7$



- Describe the error that was made in the construction of a perpendicular bisector.

Answers will vary. When drawing arcs to locate point C, the compass width was not kept the same when drawing the arcs from point A and point B. The same error was made when locating point D.



If students struggle with finding the error, have the students use a compass to try and replicate the arcs. They will find that the compass width did not remain the same when locating point C or when locating point D. Question students on why it is important that the compass width stays the same. (Answer: the compass width maintains equal distance from both endpoints of the line segment).

1.11.7 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



1.11.7 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.
Answers will vary.



The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

1.12 Angle Bisectors

Lesson Introduction

 Benchmark	MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.		 Learning Target	I can construct an angle bisector.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is an angle bisector? - How can an angle bisector be used to determine measures?			
 Lesson Narrative	This lesson explores what an angle bisector is and how to construct one using a protractor, patty paper, or compass and straightedge. Students will solve for variables and angle measures using their understanding of angle bisectors.			
 Component Summary	1.12.1 Warm-Up (~5 min.)	Explain agreement or disagreement with student statements (<i>SE p. 75</i>)	1.12.3 Guided Instruction (15 min.)	Solving for variables and angle measures using an angle bisector (<i>SE p. 78</i>)
	1.12.2 Collaboration (20-25 min.)	Constructing an angle bisector (<i>SE p. 76</i>)	1.12.4 Wrap-Up (~5 min.)	Constructing an angle bisector (<i>SE p. 79</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> Angle Bisector <u>Additional Terms:</u> N/A		- Protractor - Compass - Ruler - Patty Paper	
			Consider making large copies of the construction steps that can be moved around and put in order on the board during the whole-class discussion of the answer.	

 Teacher Tips	Common Misconceptions - Students may misread the protractor during 1.12.2 Collaboration. - Students may not open the compass width wide enough during the construction.	Things to Consider Consider how to help students remember the different steps to the constructions in the unit. Students may easily forget steps if they do not understand the whys, the geometric definitions, or the ideas that resulted from the constructions. Consider asking students to write down the four different constructions in the unit and what they remember about each. Use this information to support student learning as they prepare for any unit assessments.
	Literacy and Language Routines Discussion Supports During 1.12.2 Collaboration, students will be able to discuss and defend a variety of approaches for finding an angle bisector. It may be interesting to ask students to have a mini debate regarding which method is best. Do not clarify if best means most efficient or most accurate; allow students to discuss the merits of each method.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.1.1: Participate in Effortful Learning Starting with the 1.12.2 Collaboration, students are provided with a number of opportunities to discuss and explore what an angle bisector is. Students should be encouraged to work with a partner and discuss options and other ways to explore the angle bisectors.
	 Differentiate Instruction Using the strategy of a silent board game during 1.12.2 Collaboration question 4 allows for different entry-points for learners. Students choose when to go to the board to make a move or give an answer based on when they feel comfortable with their knowledge, instead of being selected based on other criteria.	
Lesson Notes:		

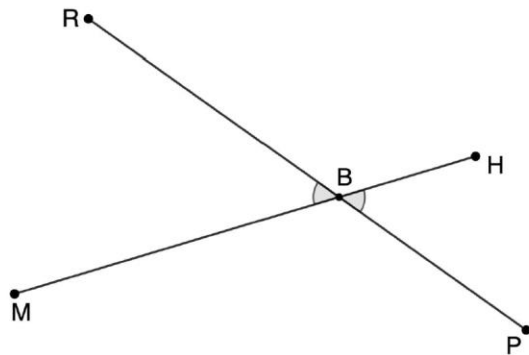
1.12.1 Warm-Up: Agree or Disagree

Component Overview: Students will engage in an agree or disagree activity where they will review the statements of two students to determine who they agree with or disagree with.



1.12.1 Warm-Up: Agree or Disagree

1. Line segments MH and RP intersect at point B .



For this activity, consider having students move to one side of the room if they agree with Jamessia, and the other side with Malik. If they don't agree with either, have them remain seated. Allow students to share out their rationale.

- a. Jamessia says $\angle MBR$ and $\angle PBH$ have the same measure. Malik disagrees and says that obviously the measure of $\angle MBR$ is larger.

Where do you stand?

- ☐ I agree with Jamessia.
- ☐ I agree with Malik.
- ☐ I disagree with both students.

- b. Explain your choice for Part A.

Answers will vary. $\angle MBR$ and $\angle PBH$ are vertical angles so they must have equal measure.

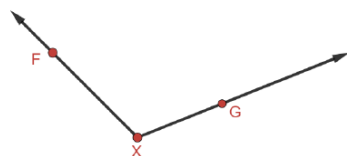
1.12.2 Collaboration: Bisecting an Angle

Component Overview: Students will find the bisector of an angle using a protractor and using patty paper. Then, the students will construct the bisector of an angle using a compass and straightedge.

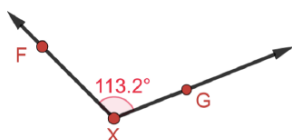


1.12.2 Collaboration: Bisecting an Angle

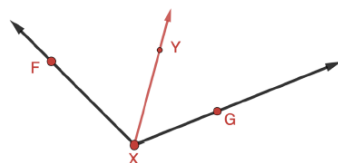
1. $\angle FXG$ is shown.



- a. Measure the angle with a protractor.
Answers will vary. Answers should be close to 113° but do not have to be exact.

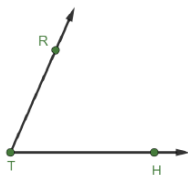


- b. Draw $\overrightarrow{XY}$ that bisects $\angle FXG$.



- c. Discuss with your partner why bisecting $\angle FXG$ using a protractor and a ruler may not be accurate. Summarize your discussion.
Answers will vary.

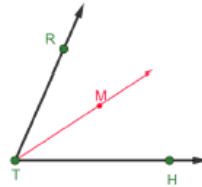
2. $\angle RTH$ is shown.



This collaboration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. Monitor while students are working to determine if there are any questions that need to be debriefed before moving to the next component. This could include highlighting student voices to share rich discussions or comments overhead while you circulated or to address common misconceptions that were evident among the majority of the class. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed collaboration.

1.12.2 Collaboration: Bisecting an Angle (continued)

- Use patty paper to copy $\angle RTH$.
- Fold the patty paper so that $\overrightarrow{TR}$ and $\overrightarrow{TH}$ lie on each other. Open the paper and draw $\overrightarrow{TM}$ along the crease.



- Explain how folding the patty paper bisects the angle.

Answers will vary. When $\overrightarrow{TR}$ and $\overrightarrow{TH}$ line up, the distance from the fold is the same, so the angles are equal. The angles being equal is what makes $\overrightarrow{TM}$ a bisector of the angle.



After question 2c, consider asking the following:

- What transformation is represented in this step?
- What maintains congruence when using this transformation?

Another way to more accurately bisect an angle is to use a compass and a ruler.

- $\angle PGW$ is shown.

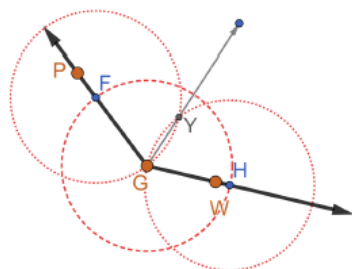
Work with your partner and use the steps below to bisect $\angle PGW$.

- Place the compass' point on point G .
- Adjust the compass' width to any width that will stay on the angle.
- Use the setting to draw an arc that extends across each leg of $\angle PGW$. Label the intersections of the arc and the two sides of the angle points F and H .
- Place the compass' point on point F . Keep the width of the compass the same or open the compass wider than the width used to draw the arc in the previous step. Draw an arc in the interior of the angle.
- Without changing the compass' setting, place the compass' point on point H and make an arc in the interior of the angle. The two small arcs in the interior of $\angle PGW$ should be intersecting. If they are not intersecting, adjust the compass to a larger width and draw new arcs from points F and H .



1.12.2 Collaboration: Bisecting an Angle (continued)

- Draw and label point Y where the two arcs intersect.
- Use a straightedge to draw a ray from point G through point Y .



$\overrightarrow{GY}$ bisects $\angle PGW$.

4. Olivia found some steps for bisecting an angle on the internet. She wrote them down and then mixed them up to test herself. Her mixed-up steps are shown. Work with your partner to order the steps 1-9.

9	Draw a ray from the vertex of the angle to the intersection of the two small arcs.
2	Stretch the compass to any length that stays on the angle.
7	Without changing the span on the compass, place the point of the compass on the other point made by the arc and the other side of the angle and make a similar arc.
4	Place the compass point on one of the two new points made by the arc.
5	Change the compass width so that the pencil stays between the rays of the angle.
1	Place compass point on the vertex of the angle.
3	Swing the compass to make an arc that crosses both rays of the angle.
6	Draw an arc in the interior of the angle.
8	Mark the point where the two small arcs in the interior of the angle intersect.



After partners have worked to order the steps, consider implementing a silent board game. With large copies of the steps prepared ahead of time, have students volunteer to come to the board to place the steps in order. Each student will move only one step at a time. Continue allowing students to come until all steps are in order and all agree on the steps. There is no talking during this activity. This activity provides different entry-points, as students volunteer when they know a particular step even if they do not know all of the steps. Using this teaching strategy now will set up expectations if it is used in other lessons, particularly when writing proofs.

1.12.3 Guided Instruction: Using Angle Bisectors to Solve Problems

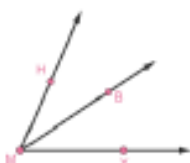
Component Overview: Students will use the information about angle bisectors to solve for unknown variables and angle measures. Students will also examine the thinking of two sample students' work and write a note to the student who made a mistake.



1.12.3 Guided Instruction: Using Angle Bisectors to Solve Problems

An **angle bisector** is a line, ray, or line segment that cuts an angle into two equal parts.

1. $\angle HMX$ is shown where $\overrightarrow{MB}$ bisects $\angle HMX$.

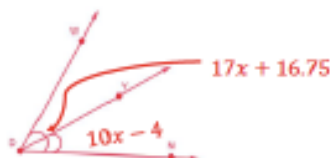


If $m\angle HMX = 7x + 3$ and $m\angle BMX = 3x + 6$, then find the value of x .

$x = 9$

2. $\overrightarrow{DY}$ bisects $\angle WDN$ such that $m\angle YDN = 10x - 4$ and $m\angle WDN = 17x + 16.75$.

- a. Draw a diagram of the description.



- b. Yordis and Hae-Won used the given information to determine the value of x . Their equations are shown.

Yordis	Hae-Won
$10x - 4 = \frac{1}{2}(17x + 16.75)$	$\frac{1}{2}(10x - 4) = 17x + 16.75$

- c. Discuss with your partner which student made a mistake setting up the equations.

Answers will vary. Hae-Won should multiply $10x - 4$ by 2 not $\frac{1}{2}$.



Some students may benefit from a discussion on the difference between $m\angle HMX$ and $\angle HMX$. It may also be helpful for students to know that the ray $\overrightarrow{MB}$ and line segment $\overline{MB}$ can both be considered the bisector.

1.12.3 Guided Instruction: Using Angle Bisectors to Solve Problems (continued)

- d. Write a short note to the student who made a mistake in setting up the equations to explain their error to them.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.

3. $\overrightarrow{BT}$ bisects $\angle HBN$ such that $m\angle HBN = 10x - 2$, $m\angle HBT = 5x - 1$, and $m\angle TBN = 7x - 23$. Find $m\angle TBN$.
 $m\angle TBN = 54$



Before moving to question 2d, have students share a summary of their discussion. Encourage students to talk about which angle is the larger and which angle is smaller. Ask how multiplying by $\frac{1}{2}$ affects the value of a number. When a student gives a valid explanation of which student is correct, have another student repeat what was said. Doing this here will help all students organize their thoughts to write an explanation of the error in the following question.



Question 3 can be solved in several ways. Consider releasing students to try this on their own. Monitor for the strategies used and call on students to share their thinking with the group as an opportunity to highlight different ways of thinking about the same problem.

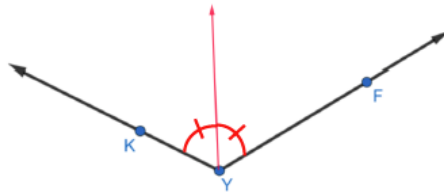
1.12.4 Wrap-Up: Ticket Out the Door

Component Overview: Students will construct the bisector of an angle.



1.12.4 Wrap-Up: Ticket Out the Door

1. Construct the angle bisector of $\angle KYF$ using a compass and a straightedge.



Consider allowing students to construct the angle bisector or to explain how to construct the angle bisector. Both options provide information about what students understand from the lesson.